Course Name: *Advanced Underground Mining Methods*
Course Code: *MND 411*

# Longwall Mining

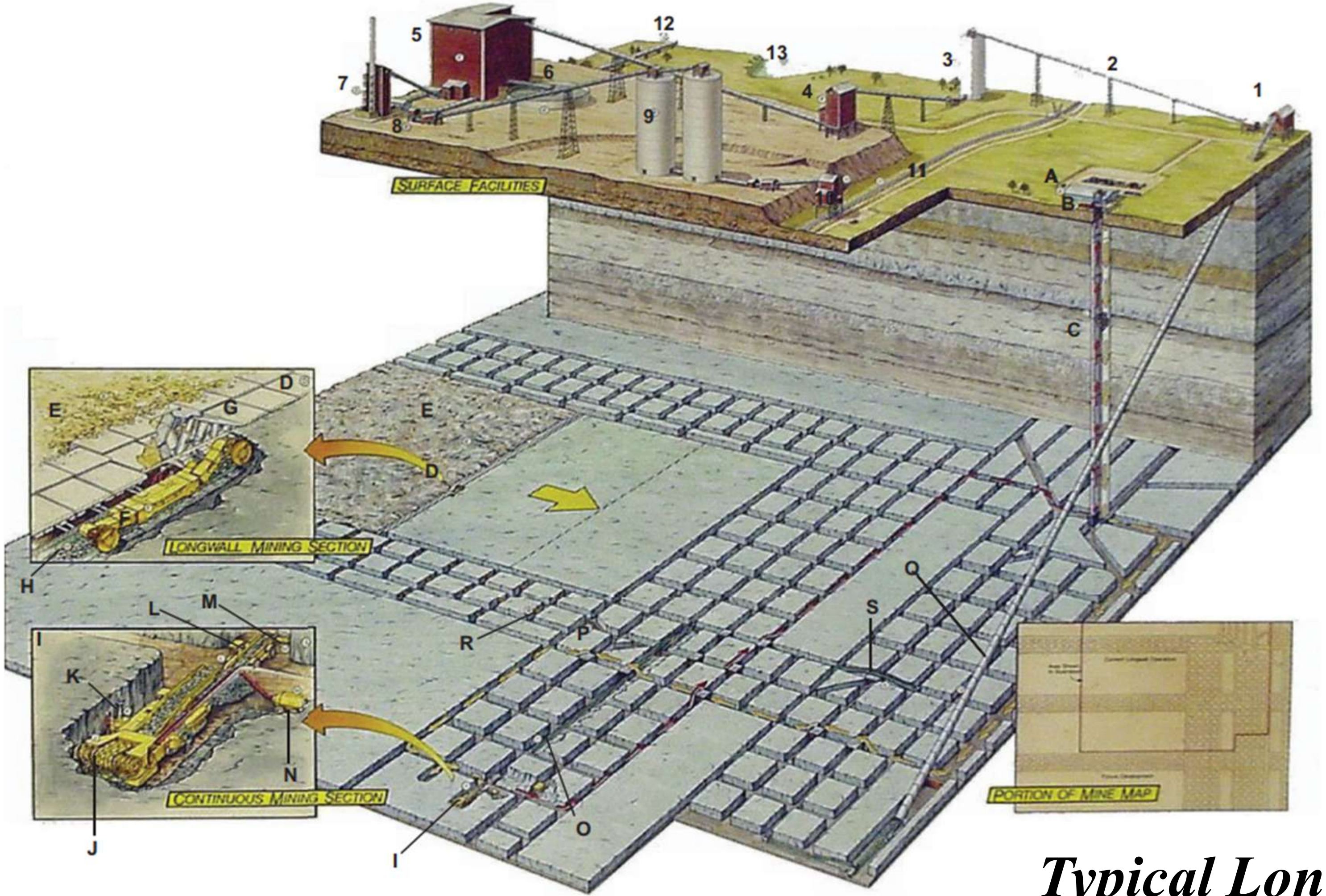
## *Introduction:*

- Longwall method of mining consists of laying out **long straight face** either **along the dip-rise direction** or **along the strike direction** of the coal seam from which all **the coal is removed in one working section** in a **series of operations** maintaining a **continuous line of advance in one direction** and leaving behind the **void** which may be **caved or stowed** with sand.

- Longwall mining consists of formation of a large pillar by driving roadways called gate roads.

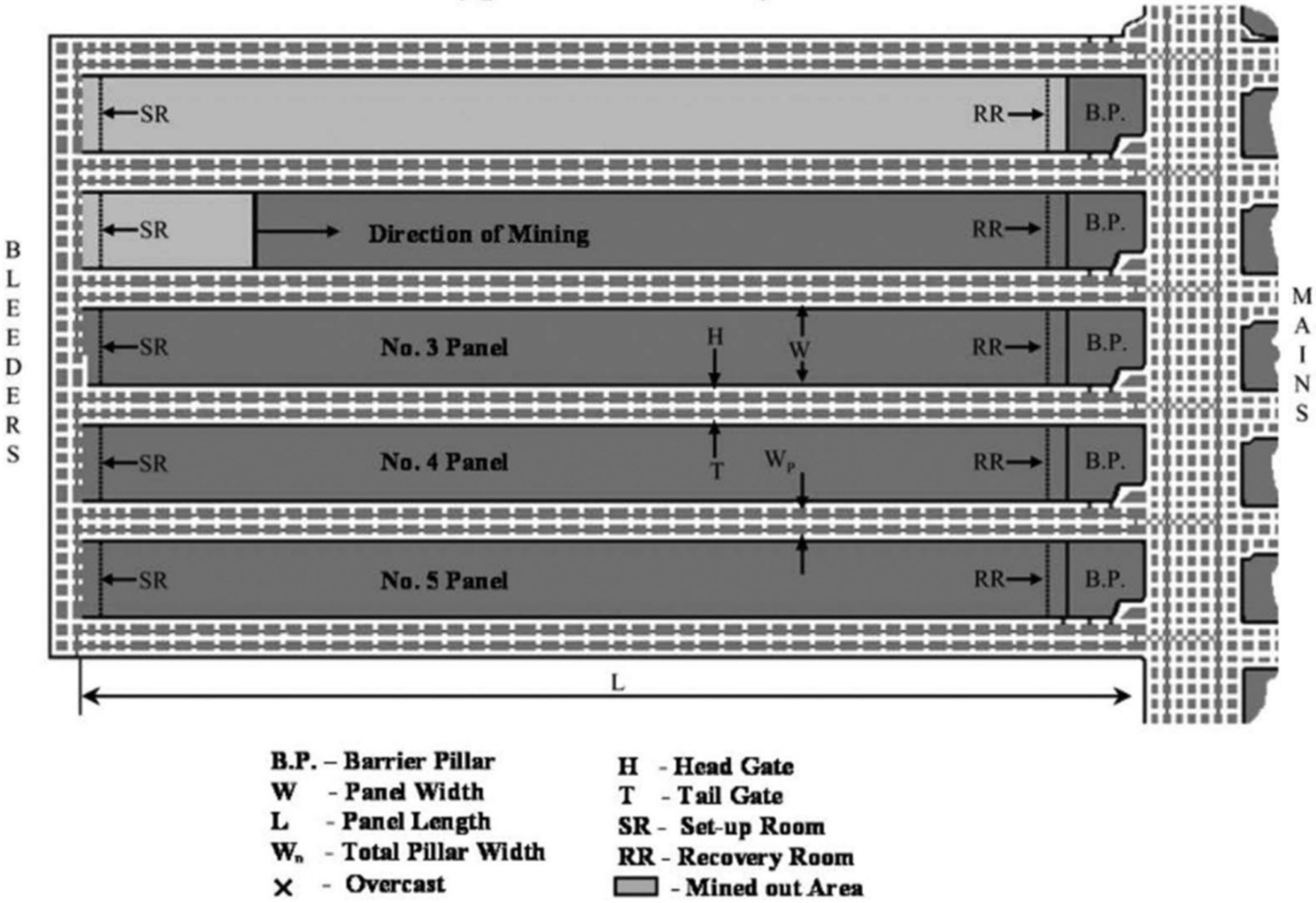
- The face length may vary from 80m to 300m and panel length from few hundred metres to 2km depending on the availability of area/reserves available for longwall mining.

# Introduction:




**Typical Longwall Panel Layout**

# Introduction:




***<u>A typical panel layout of a longwall mining section</u>***

# *Applicability:*

## ***1. Reserve:***

- A large coal reserve is needed to make the investment feasible.

## ***2. Geology:***

- Geology is the single most important factor in high production, high efficiency longwall mining.
- The geological factors affecting panel layout include ***seam characteristics***, ***roof and floor strata***, and ***geological anomalies***

## ***Seam Characteristics***

✓ **Gradient of the seam**

- Seams dipping up to 1 in 3 can be easily extracted by longwall method. For steeper gradient, special methods like inclined slicing or horizontal slicing method can be used.

# *Applicability:*

## ***Seam Characteristics***

✓ **Seam thickness**

- Coal seams having thickness range from 0.6 – 6.0m can be effectively worked in one section with powered support and shearer/plough.
- Thicker seams (>6m) can be worked in slices of 2-3m thick, either in ascending or descending order.
- The thinnest seam that can be worked is about 0.4m only depending on the quality of coal.
- The thickness of the seam should be more uniform throughout the panel for effective mining by longwall method.

## *Applicability:*

### ***Roof and Floor strata***

***Caving behaviour of strata***

- Roof should be easily cavable. Consistently safer and productive operation of longwall faces is very difficult in adversely caving strata condition unless proper planning and reasonable solution are not readily available for timely management of the impending problems.

***Influence of floor strata***

- Floor Heaving and Puncturing depend on the strength of the floor and the amount of load transfer to the floor through face and roof characteristics

## *Applicability:*

### ***Geological Anomalies***

- Geological conditions that affect production, and thus panel layout, include faults, folds, sandstone channels, clay veins, and fractures (mountain seams or hill seams).

### ***3. Cover depth***

- Longwall method is the best method for underground exploitation of coal at depth more than 200m from strata control as well as ventilation points of view.

### ***4. Panel Dimension:***

- The general concept is to design the panels to be as large as possible, mainly to reduce the development work.
- Currently the average panel width in 2019 is about 360 m, and the maximum width is 482 m. The average panel length is about 3459 m, and the maximum length is 6860 m.

## *Applicability:*

### ***5. Strength of coal***

- Now a day, with increase in available machine power, the strength of coal has little importance. However, mechanized longwall with shearer or plough demand soft or medium hard coal for better workability.

### ***6. Gassiness of the seam***

- Highly gassy coal seams can also be worked safely maintaining effective level of ventilation which may not be possible in other methods. As the method of coal winning is strictly based on mechanical cutting, hazards associated with use of explosive is not there.

## *Advantages of Longwall Method:*

- Potential of high production and productivity
- A longwall face can produce from 2000-10000t per day depending on seam thickness and type of mechanization
- Better ventilation and lesser chance of fire and spontaneous heating
- High percentage of extraction, almost 100% within the panel
- Better supervision and control of operations due to concentrated working at one place
- Better roof control due to more systematic sequence and line of extraction
- High productivity and less cost of production
- Gassy seams and seams liable to spontaneous heating can be worked with least risk
- Wide application under different conditions with respect to thickness, gradient and depth

## *Disadvantages of Longwall Method:*

- High capital investment
- High gestation period
- Can not be applied in geologically disturbed areas, therefore the investment decision is subjected to high risk.
- High amount of subsidence observed at surface
- The method needs highly multi skilled manpower.
- The installed power requirement of a longwall panel is very high requiring dedicated power distribution

## *Longwall Advancing Vs. Longwall Retreating*

- Longwall advancing
  - In longwall advancing system, a longwall face (100-150m long) is advanced away from the main transport route either in the dip direction or in the strike direction simultaneously with the gate roads.
  - In this system, gate roads are advanced simultaneously with the face advance. The gate roads are kept 5m advanced from the face and duly supported as the face advances.

# *Longwall Advancing Vs. Longwall Retreating*

➢ Longwall retreating

- In longwall retreating system, gate roads are driven from the main transport route upto the panel boundary and then, they are connected forming a longwall face, which is retreated from the panel boundary towards the main transport route.
- In this system, the gate roads are supported by solid coal pillars and can be abandoned as the face retreats.

## *Longwall Advancing Vs. Longwall Retreating*

- Advantages of longwall advancing

- Fully productive operation of longwall mining can be started with little development work.
- It does not require that the gate roads need to be driven first up to the panel boundary as in case of retreat fashion. Thus, gestation period is less.
- No investment and separate organisation is necessary for the drivage of the gate roads.

# *Longwall Advancing Vs. Longwall Retreating*

- Disadvantages of longwall advancing

- The gate roads need to be maintained throughout the life of the panel in the extracted area or through goaf. It requires heavy support which is difficult to maintain.
- A much rigid cycle of operation needs to be followed. The gate roads needs to be simultaneously advanced along with the advance of the longwall face.
- The rate of face advance depends on the rate of advance of the gate roads, higher output can not be expected.

## *Longwall Advancing Vs. Longwall Retreating*

- Geology of the area can not be explored in advance as it can be known in retreating system.
- Drivage of gate roads are to be done with drilling and blasting. Fast heading machine like road header or dint header can not be used.
- As the gate roads are driven along the face, more dust is carried onto the face.
- Ventilation of the face is poor compared to retreat mining as much air is leaked through the goaf in advancing system

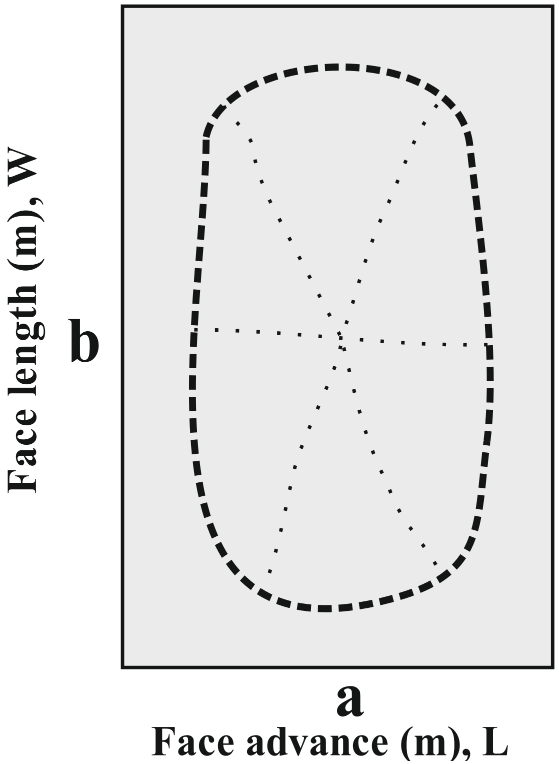
## *Different Parameters of Longwall Mining*

- Individual variants of longwall mining include:
  - Length of the face
  - Size of chain pillar
  - Rate of advance
  - Orientation of face, either along strike or along dip-rise
  - Method of coal winning, drilling and blasting, cutting by shearer or plough
  - Type of face support
  - Method of roof control, caving or stowing

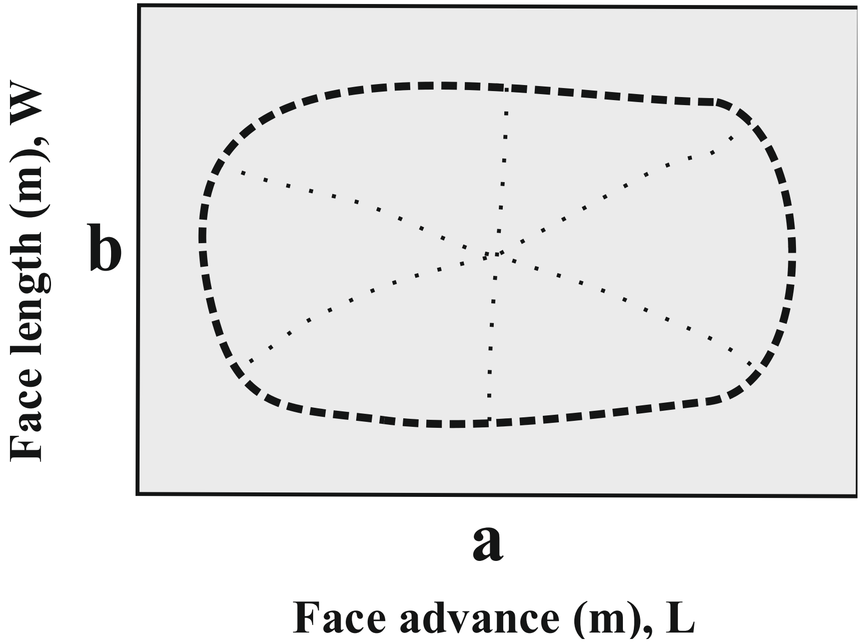
## *Different Parameters of Longwall Mining*

- Length of longwall face
  - The length of a longwall face is a problem of optimisation and it depends on lot of factors.
  - Variation in any of the factors affect the other.
  - It is a common understanding that higher the length of longwall face, higher is the production. But it is not always true because production from a longwall face depends on many other factors while the geo-technical and rock-mechanics properties of the locale governs the technical feasibility of longwall face length.
  - The overall economics has a decisive role on the face length.

# *Different Parameters of Longwall Mining*




Case1: a/b<1




Case 2: a/b>1

## *Different Parameters of Longwall Mining*

- The major factors which govern the choice of longwall face length are as follows:
  - Strata control
  - Capacity of longwall face equipment
  - Strata characteristics
  - Production and productivity
  - Ventilation at the face
  - Human factors
  - Investment and cost of production

# *Different Parameters of Longwall Mining*

- Strata control
  - Higher the face length, higher is the support requirement as the exposed area increases
  - Depending on the roof condition, the stability of working will be limited.
  - From the roof control point, the face length should be restricted to 150m or so.
  - The pressure at the middle of the face increases as the face length increases.
  - Unless otherwise restricted to geological disturbances like fault, dykes etc or physical or statutory limitations due to fire or water bodies, the maximum length of longwall face is not defined.

## *Different Parameters of Longwall Mining*

- Capacity of longwall face equipment
  - Due to technical limitation of face conveyor design, face exceeding 300m length is not feasible.
  - Present instruments are capable of operating maximum up to the length of 150-250m beyond which reliability of the machine goes down appreciably, requiring very strict maintenance schedule.
  - Longer conveyor needs high capacity motors which may not be feasible to install in the restricted area of the longwall face.

## *Different Parameters of Longwall Mining*

- Strata characteristics
  - When the immediate roof is unstable, or where the seam is liable to burst, the face is prone to undergo spalling
    - In all these conditions, fast rate of face advance is desirable. In such cases, operating a smaller length of face is more advantageous to shorten the duration of passing the disturbed area.

# *Different Parameters of Longwall Mining*

- Production and productivity
  - Although it is well established that production and productivity increases with face length, it has also a direct effect on capital investment and cost of production
  - Ventilation at the face
    - Longwall face creates more dust, which in turn require more amount of air for its acceptable dilution and clearance, which may not be practically feasible in all cases.

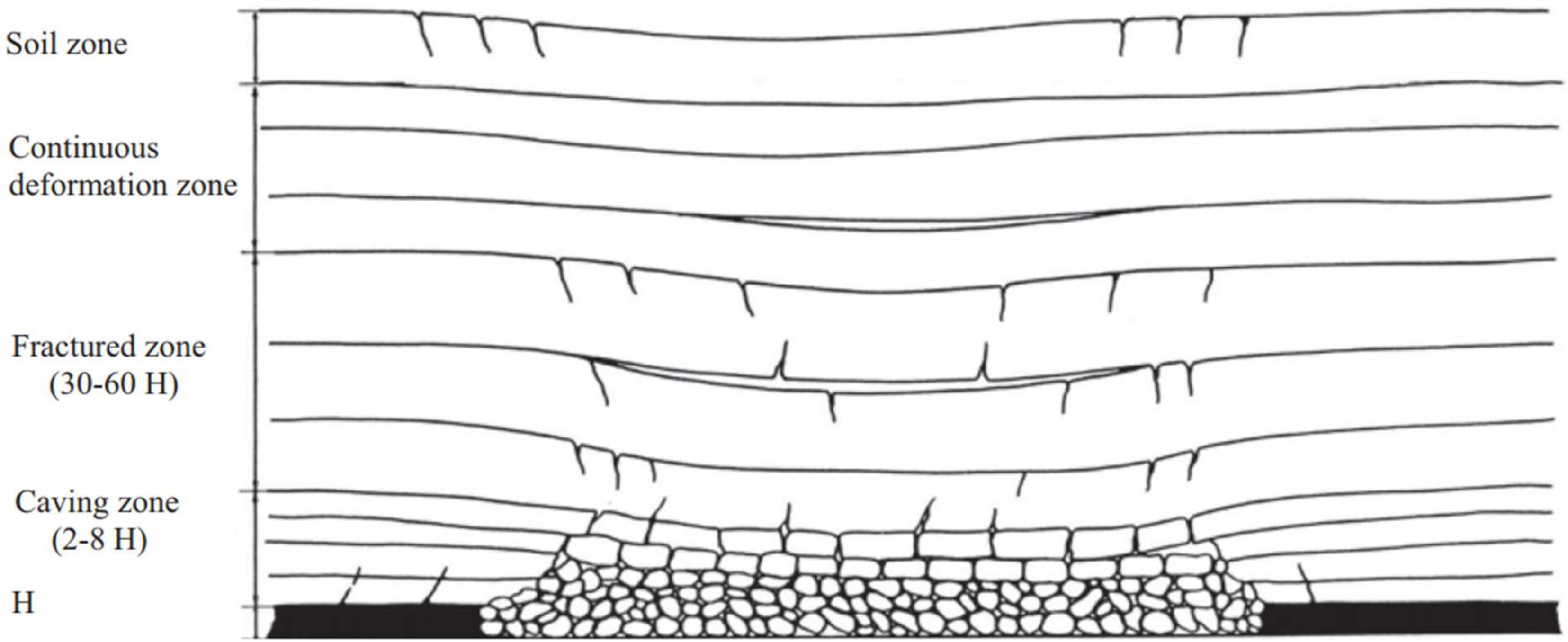
## *Different Parameters of Longwall Mining*

- Human factors
  - Longer face creates more fatigue

- Investment and cost of production
  - Higher the face length, higher is the production and the productivity, but higher is the investment required for this purpose. Thus, an optimization has to be made among face length, capital investment and cost of production.

# *Overburden Movements*

- When a longwall panel of sufficient width and length is excavated, the overburden roof strata are disturbed in order of severity from the immediate roof toward the surface.



## ***Caving Zone***

- The caved zone, which is the immediate roof before it caves, ranges in thickness from two to eight times the height of extraction (or mining height).
- In this zone, the strata fall on the mine floor and, in the process, are broken into irregular but platy shapes of various sizes. They are crowded in a random manner. Thus, the rock volume in its broken state is considerably larger than that of the original intact strata.
- The volume ratio of broken strata to its original intact strata is called the expansion ratio or bulking factor. Expansion ratio is a very important factor because it determines the height of the caved zone.

# *Overburden Movements*

## ***Fractured Zone***

- Above the caved zone is the fractured zone. In this zone, the strata are broken into blocks by vertical and/or subvertical fractures and horizontal cracks due to bed separation.
- The adjacent blocks in each broken stratum still maintain contact either fully or partially across the vertical or subvertical fractures. Thus, there is a horizontal force that is transmitted through and remains in these strata.
- The thickness of the fractured zone ranges from 28 to 52 times the mining height. The combined thickness of the caved and fractured zones ranges from 30 to 60 times the mining height.

# *Overburden Movements*

## ***Continuous Deformation Zone***

- Between the fractured zone and the surface is the continuous deformation zone.
- In this zone, the strata deform gently without causing any major cracks that extend long enough to cut through the thickness of the strata, as in the fractured zone. Therefore, the strata behave essentially like a continuous or intact medium.

# *Caving Roofs:*

## ***Immediate Roof***

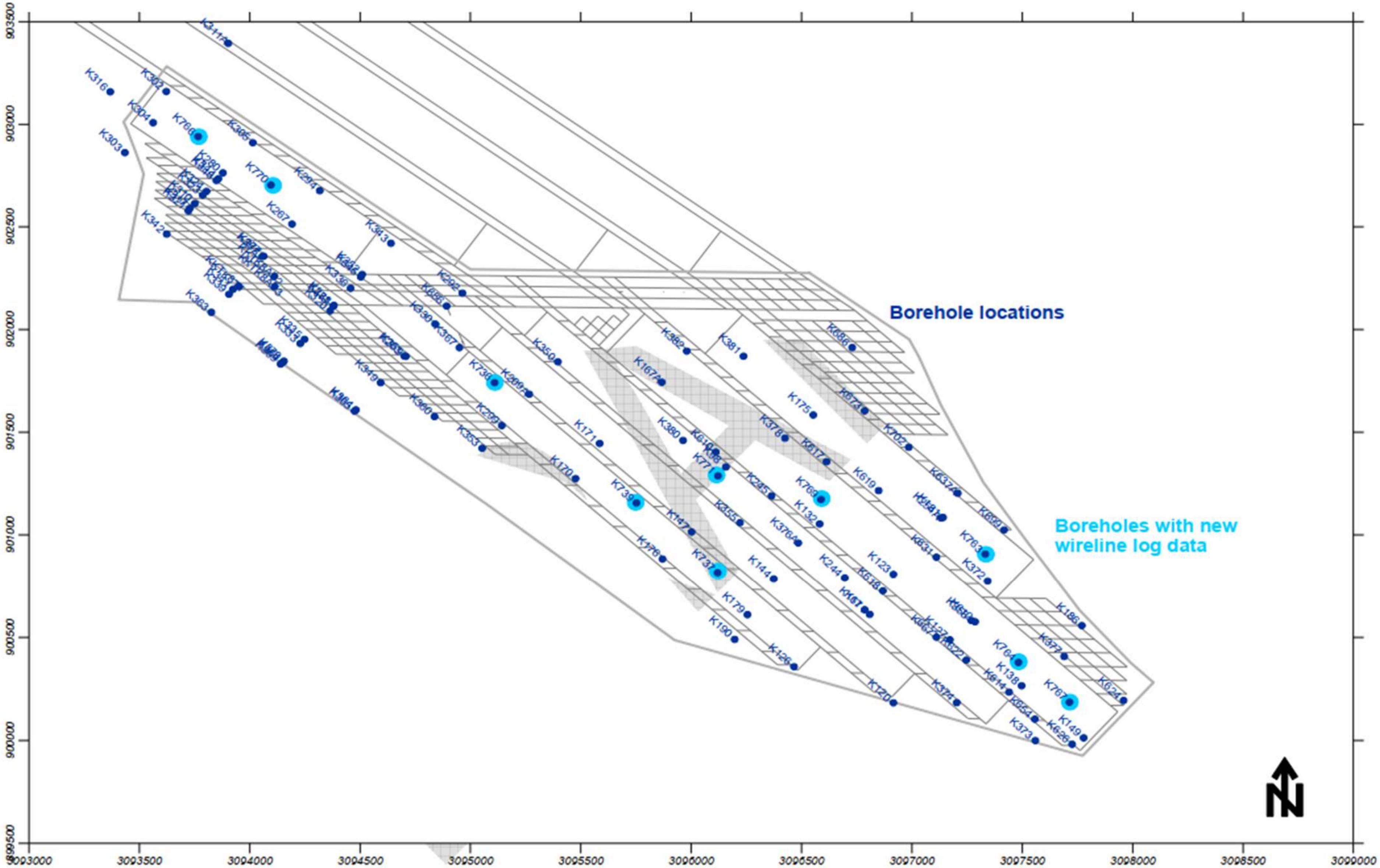
- The phrase "**immediate roof**" has been used in this work for a layer just above the coal seam to be extracted.
- It is amenable to cave after advance of the support if it is weak and thin or sufficiently laminated.
- If it is strong and difficult to cave, it overhangs to a considerable length in the goaf and plays a vital role on loading over the support at the face.

## ***Main Roof***

- The "**main roof**" lies just above the immediate roof.
- The failure of this layer in large span, conceptually, causes loading of support at the face resulting in face convergence as well as high abutment stress during the periods of main fall and periodic caving of the strata.
- The main roof may consist of either a single layer or a series of layers which may separate from each other during progressive caving.

## *Roof Separation Index (RSI)*

- Used to identify the prominent layers of roof separation in the caving zone which has been assumed to be limited within maximum height of 15 times the extraction thickness.
- The roof layers above this height do not play any role in active caving and associated loading on the supports with progressive advance of a longwall face.
- RSI is an index containing six parameters. A high rating indicates more stable rock mass.
- The rating distribution of first three parameters namely, Uni-axial compressive strength (UCS), RQD and average core length (ACL) are similar to Bieniawski (1976) RMR as these parameters are common.
- The rating distribution of fourth parameter i.e. ground water is also considered from Bieniawski (1976) RMR, but only two extreme cases of the classification have been considered.
- The rating distribution of last two parameters i.e. bed thickness and rock type are based on field experience and engineering judgment.



Borehole locations
Boreholes with new wireline log data
N
3093000
3093500
3094000
3094500
3095000
3095500
3096000
3096500
3097000
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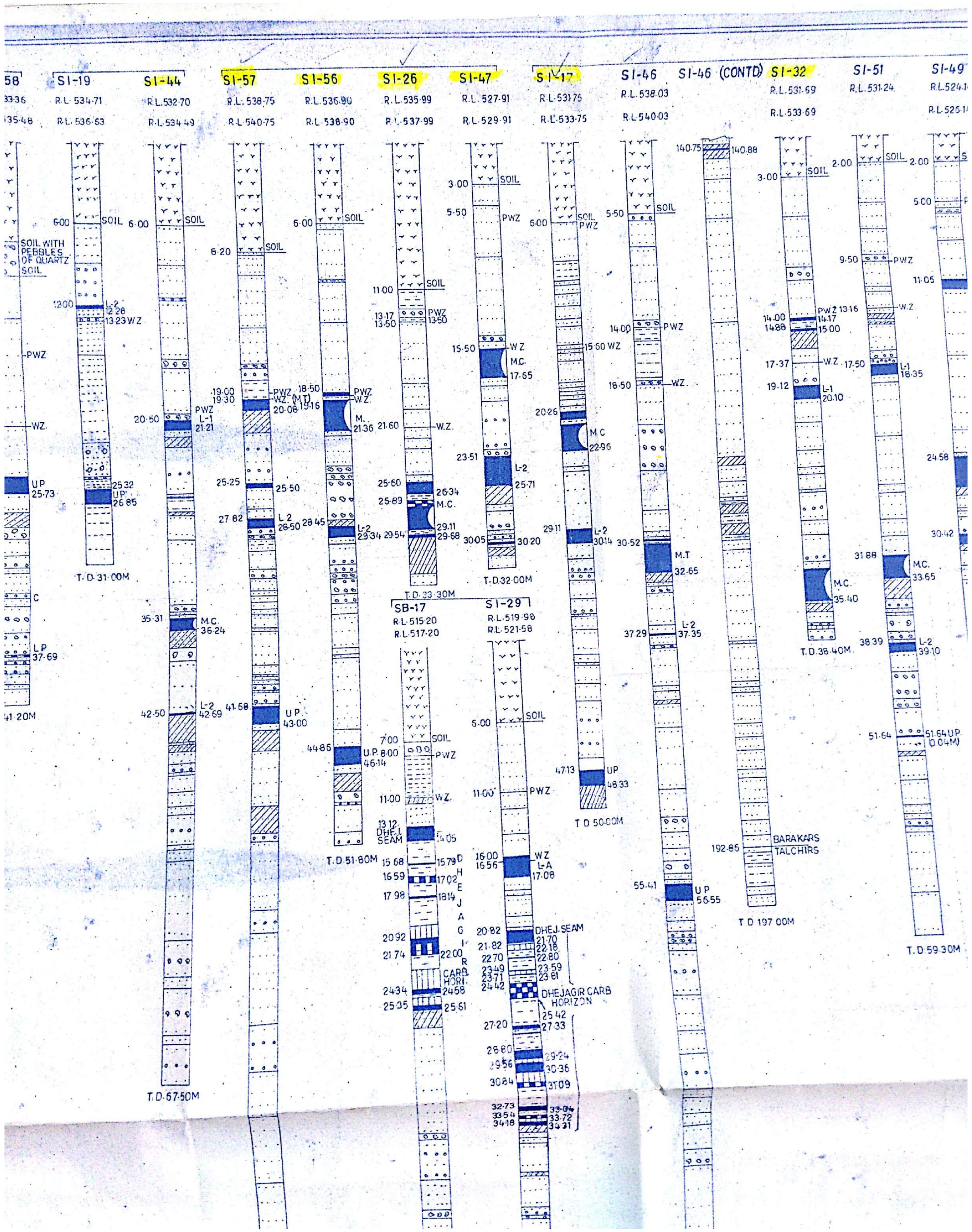

S1-19
R.L. 534·71
S1-44
R.L. 532·70
S1-57
R.L. 538·75
R.L. 540·75
S1-56
S1-26
R.L. 535·99
S1-47
R.L. 527·91
R.L. 529·91
S1-17
R.L. 531·75
R.L. 533·75
S1-46
R.L. 538·03
R.L. 540·03
S1-46 (CONTD)
S1-32
R.L. 531·69
R.L. 533·69
S1-51
R.L. 531·24
S1-49
SB-17
R.L. 515·20
R.L. 517·20
S1-29
R.L. 519·98
R.L. 521·58
SOIL WITH PEBBLES OF QUARTZ
DHEJ. SEAM
DHEJAGIR CARB HORIZON
BARAKARS
TALCHIRS
T.D. 31·00M
T.D. 67·50M
T.D. 51·80M
T.D. 33·30M
T.D. 32·00M
T.D. 50·00M
T.D. 197·00M
T.D. 38·40M
T.D. 59·30M

Roof

Roof

Roof



Floor

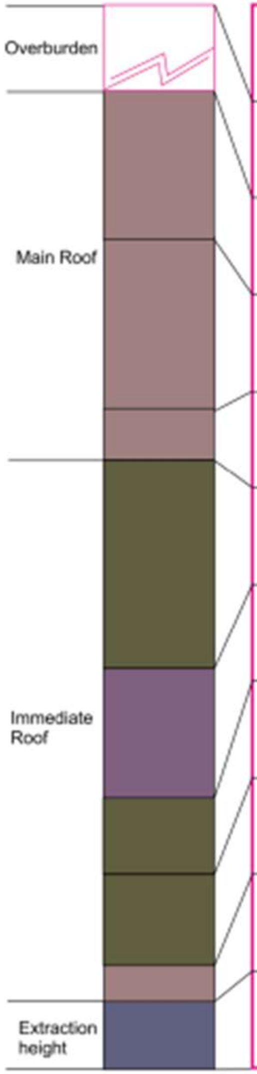
# *Roof Separation Index (RSI)*

- In order to identify the different layers in roof, different rock beds are identified as per lithology of the rock formation above the coal seam.
- The RQD, average core length and strength values are assigned to these rock beds on the basis of detailed logging results of the core samples and laboratory test.
- RSI of the rock beds are computed as an algebraic sum of the values assigned for the six different parameters. If the value of RSI of a rock bed is ≤14, it is considered that it will behave as a layer of roof separation in the series of rock beds lying above the coal seam.
- Once the roof separation layers in the overlying strata are identified using RSI, the mechanical properties of different layers in roof existing within the zone of roof caving for a given geo-mining condition are assessed.

# *Roof Separation Index (RSI)*

| Sl. No. | Parameter | Unit/ Rating | Rating distribution | | | | |
|---|---|---|---|---|---|---|---|
| 1 | Uni-axial compressive strength | MPa | <5 | 5-25 | 25-50 | >50 | |
| | | Rating | 1 | 2 | 4 | 7 | |
| 2 | RQD | % | <25 | 25-50 | 50-75 | 75-90 | >90 |
| | | Rating | 3 | 8 | 13 | 17 | 20 |
| 3 | Average core length | cm | <5 | 5-30 | 30-100 | 100-300 | |
| | | Rating | 5 | 10 | 20 | 25 | |
| 4 | Ground water condition/wetness of strata | Condition | Dry | Watery formation | | | |
| | | Rating | 10 | 0 | | | |
| 5 | Bed thickness | m | <1.5 | 1.5-2.5 | 2.5-3.5 | 3.5-4.5 | >=4.5 |
| | | Rating | -28 | -16 | -12 | -5 | 0 |
| 6 | Rock formation of bed | Rock type | Sandstone | Others (shale, coal, clay, shaly coal) | | | |
| | | Rating | 0 | -25 | | | |

# *Roof Separation Index (RSI), PVK Panel 21, SCCL*




| Thickness (m) | Rock Type | RQD (%) | Sc (MPa) | St (MPa) |
|---|---|---|---|---|
| 166.34 | Overburden | 76 | 12.5 | 1.1 |
| 7.66 | Grey and Carb.Clay | 50 | 12.5 | 1.1 |
| 8.28 | Cgsst,GW | 80 | 12.5 | 1.1 |
| 2.00 | Grey and Carb.Clay | 50 | 12.5 | 1.1 |
| 12.49 | Cg-sst,GW,Pebble | 78 | 8.7 | 0.9 |
| 6.53 | Cg-fgsst | 94 | 12.9 | 1.2 |
| 3.70 | Cg-sst,GW,Pebble | 77 | 8.7 | 0.9 |
| 4.00 | Cg-sst,GW,Pebble | 93 | 8.7 | 0.9 |
| 1.30 | Cgsst,GW | 44 | 12.5 | 1.1 |
| 3.00 | Coal | 40 | 23.1 | 1.5 |

| Rock type | Thickness (m) | RQD, % | $\sigma_c$, Mpa | $\sigma_t$, Mpa | Density, Kg/m3 | Emod, GPa | ACL, cm |
|---|---|---|---|---|---|---|---|
| Coal | 3 | 40 | 23.16 | 1.54 | 1400 | 2 | 7.70 |
| Cgsst, GW | 1.3 | 44 | 12.46 | 1.07 | 2050 | 3.86 | 8.67 |
| CGsst, GW, Pebble | 4 | 93 | 8.68 | 0.9 | 2040 | 2.69 | 24.27 |
| Cgsst, GW, Pebble | 3.7 | 77 | 8.68 | 0.9 | 2040 | 2.69 | 13.70 |
| Cg to Fg sst | 6.53 | 94 | 12.93 | 1.15 | 2060 | 4.01 | 19.2 |
| Cgsst, GW, Pebble | 12.49 | 78 | 8.68 | 0.9 | 2040 | 2.69 | 16.6 |
| Grey and carbonaceous clay | 2 | 50 | 12.46 | 1.07 | 2050 | 3.86 | 7.7 |
| Cgsst, GW | 8.28 | 80.3 | 12.46 | 1.07 | 2050 | 3.86 | 23.65 |
| Grey and carbonaceous clay | 7.66 | 50 | 12.46 | 1.07 | 2050 | 3.86 | 7.7 |
| Cgsst, GW | 8.54 | 69 | 12.46 | 1.07 | 2050 | 3.86 | 14.1 |
| Cgsst, GW | 7.3 | 85 | 12.46 | 1.07 | 2050 | 3.86 | 32.14 |
| Overburden | 150.5 | | | | | | |

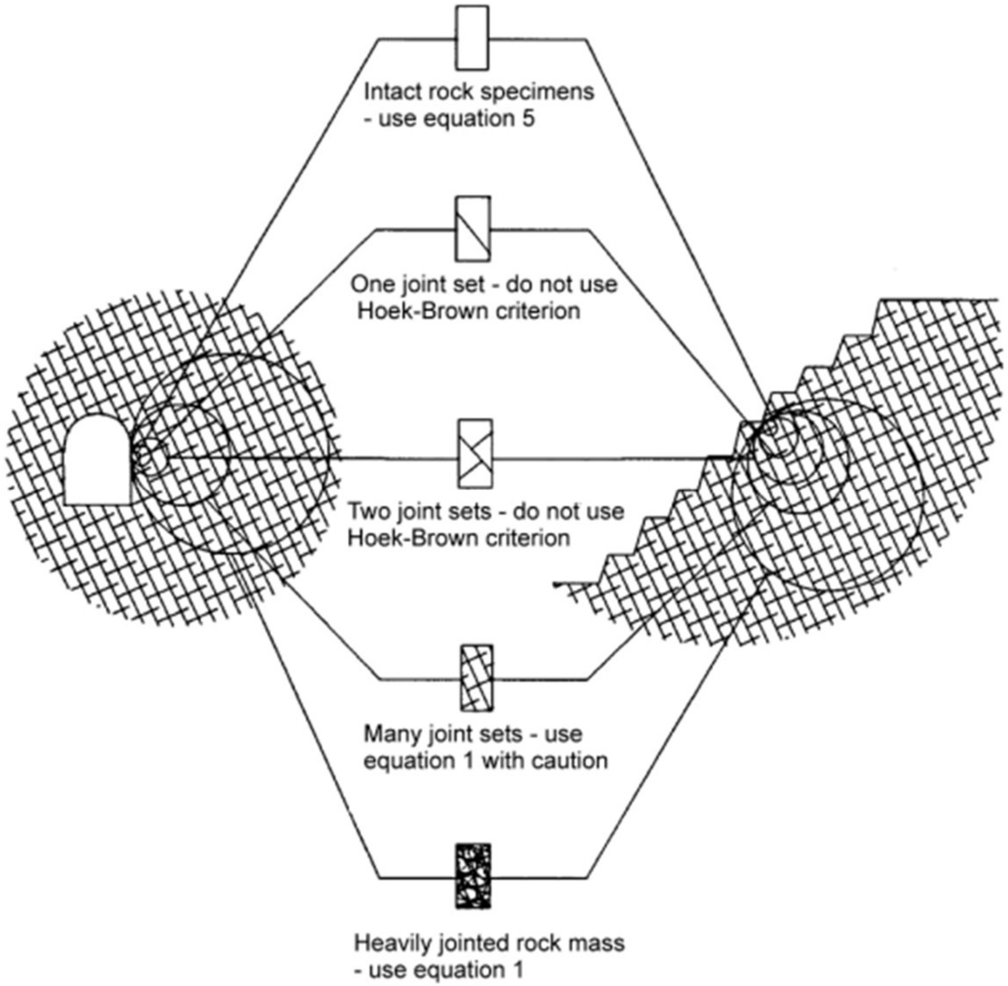
# Roof Separation Index (RSI)

| Rock type | Thickness (m) | RQD, % | $\sigma_c$, Mpa | $\sigma_t$, Mpa | Density, Kg/m3 | Emod, GPa | ACL, cm | $\sigma_c$ factor | RQD factor | ACL factor | Factor for presence of ground water | adjustment factor for thickness | adjustment factor for type of rock formation | Roof separation Index, RSI |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Coal | 3 | 40 | 23.16 | 1.54 | 1400 | 2 | 7.70 | 2 | 8 | 10 | 10 | -12 | -25 | -7 |
| Cgsst, GW | 1.3 | 44 | 12.46 | 1.07 | 2050 | 3.86 | 8.67 | 2 | 8 | 10 | 10 | -28 | 0 | 2 |
| CGsst, GW, Pebble | 4 | 93 | 8.68 | 0.9 | 2040 | 2.69 | 24.27 | 2 | 20 | 10 | 10 | -5 | 0 | 37 |
| Cgsst, GW, Pebble | 3.7 | 77 | 8.68 | 0.9 | 2040 | 2.69 | 13.70 | 2 | 17 | 10 | 10 | -5 | 0 | 34 |
| Cg to Fg sst | 6.53 | 94 | 12.93 | 1.15 | 2060 | 4.01 | 19.2 | 2 | 20 | 10 | 10 | 0 | 0 | 42 |
| Cgsst, GW, Pebble | 12.49 | 78 | 8.68 | 0.9 | 2040 | 2.69 | 16.6 | 2 | 17 | 10 | 10 | 0 | 0 | 39 |
| Grey and carbonaceous clay | 2 | 50 | 12.46 | 1.07 | 2050 | 3.86 | 7.7 | 2 | 13 | 10 | 10 | -16 | -25 | -6 |
| Cgsst, GW | 8.28 | 80.3 | 12.46 | 1.07 | 2050 | 3.86 | 23.65 | 2 | 17 | 10 | 10 | 0 | 0 | 39 |
| Grey and carbonaceous clay | 7.66 | 50 | 12.46 | 1.07 | 2050 | 3.86 | 7.7 | 2 | 13 | 10 | 10 | 0 | -25 | 10 |
| Cgsst, GW | 8.54 | 69 | 12.46 | 1.07 | 2050 | 3.86 | 14.1 | 2 | 13 | 10 | 10 | 0 | 0 | 35 |
| Cgsst, GW | 7.3 | 85 | 12.46 | 1.07 | 2050 | 3.86 | 32.14 | 2 | 17 | 20 | 10 | 0 | 0 | 49 |
| Overburden | 150.5 | | | | | | | | | | | | | |

# *Roof Separation Index (RSI)*

| Layer | Thickness, m | Density, $Kg/m^3$ | RQD, % | Comp strength, MPa | Tensile strength, MPa |
|---|---|---|---|---|---|
| Coal | 3 | 1400 | 40 | 23.16 | 1.54 |
| Immediate roof layer | 30.02 | 2045 | 80 | 10.02 | 0.97 |
| Main roof layer | 15.94 | 2050 | 66 | 12.46 | 1.07 |
| Overburden | 166.34 | 2050 | 76 | 12.46 | 1.07 |

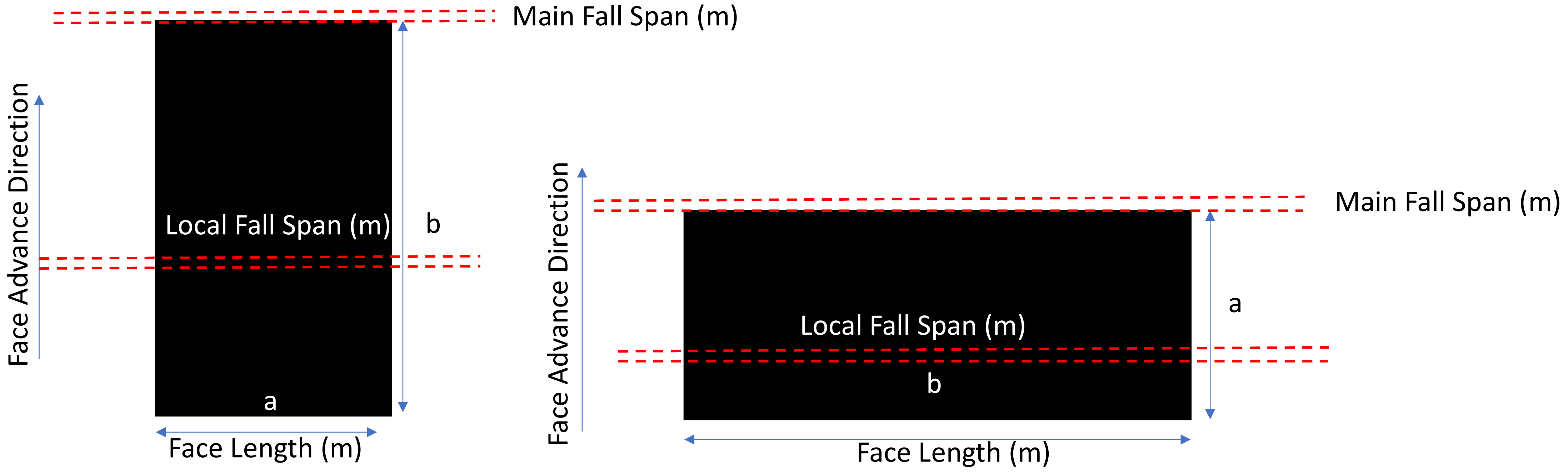
# *Conversion of Intact to Rock Mass Properties*

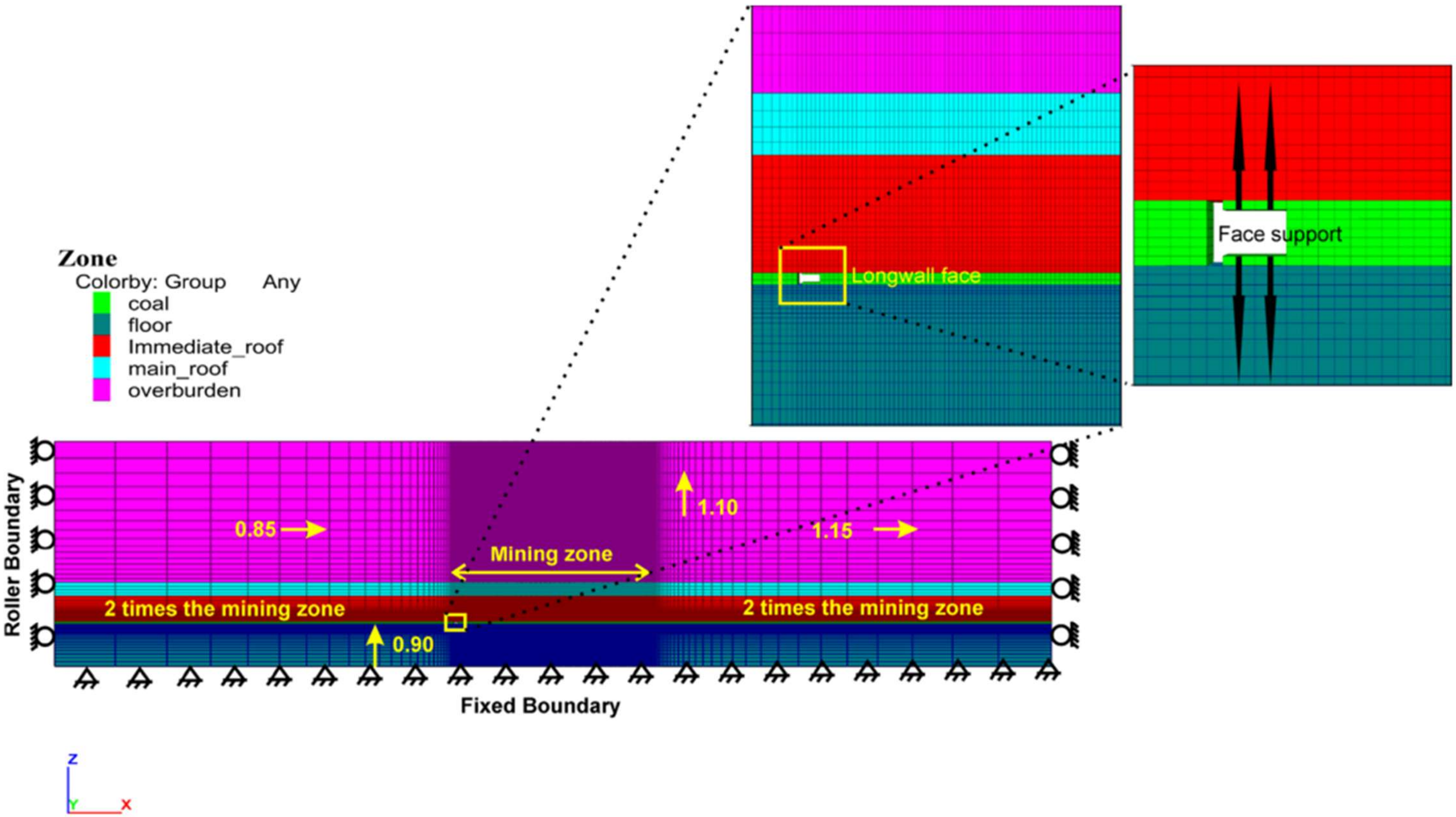


# *Estimation of Caving Span*

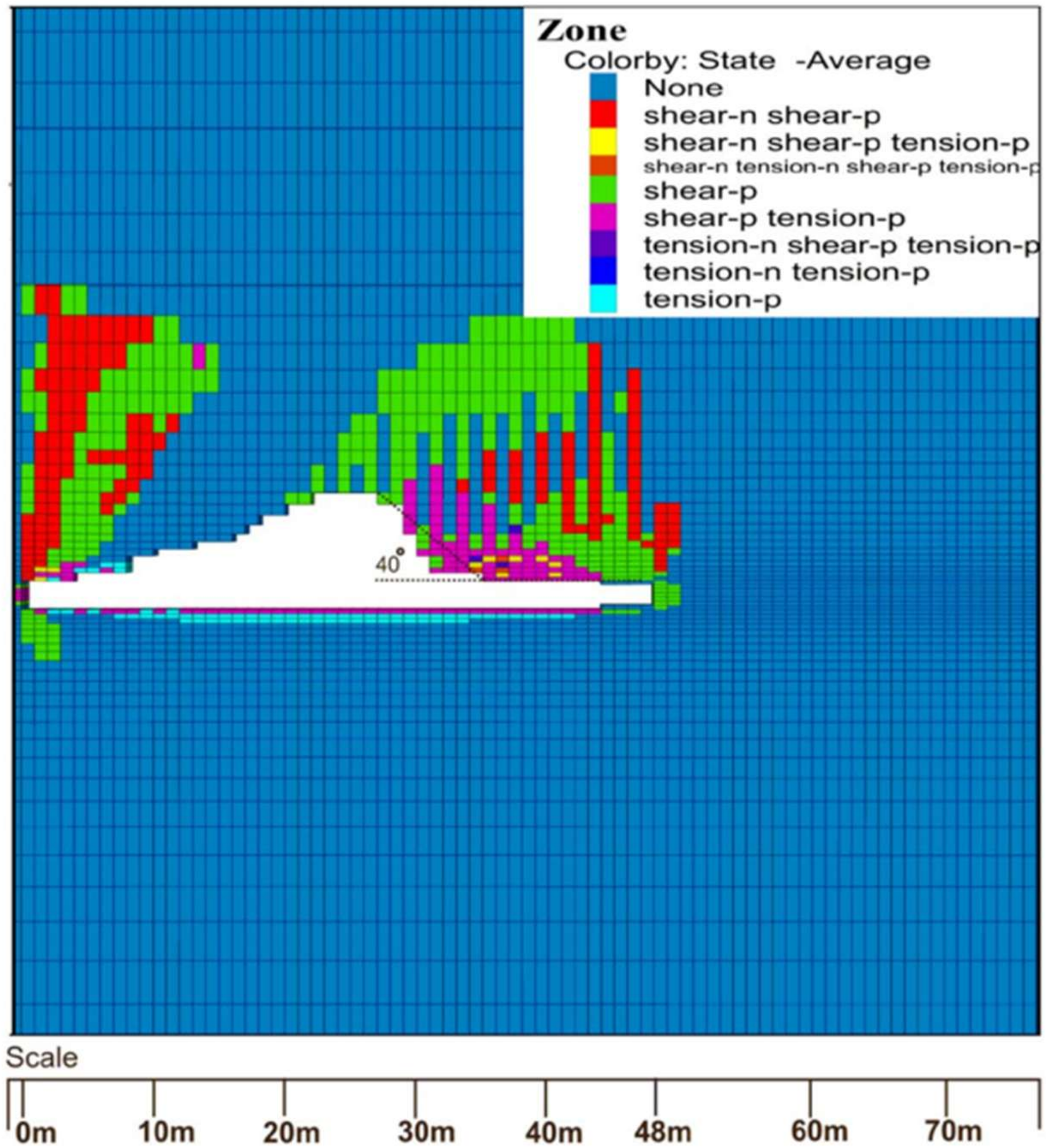
## Beam and Cantilever Model

- Applicable when one dimension of the panel is sufficiently larger than another, i.e. b/a > 2



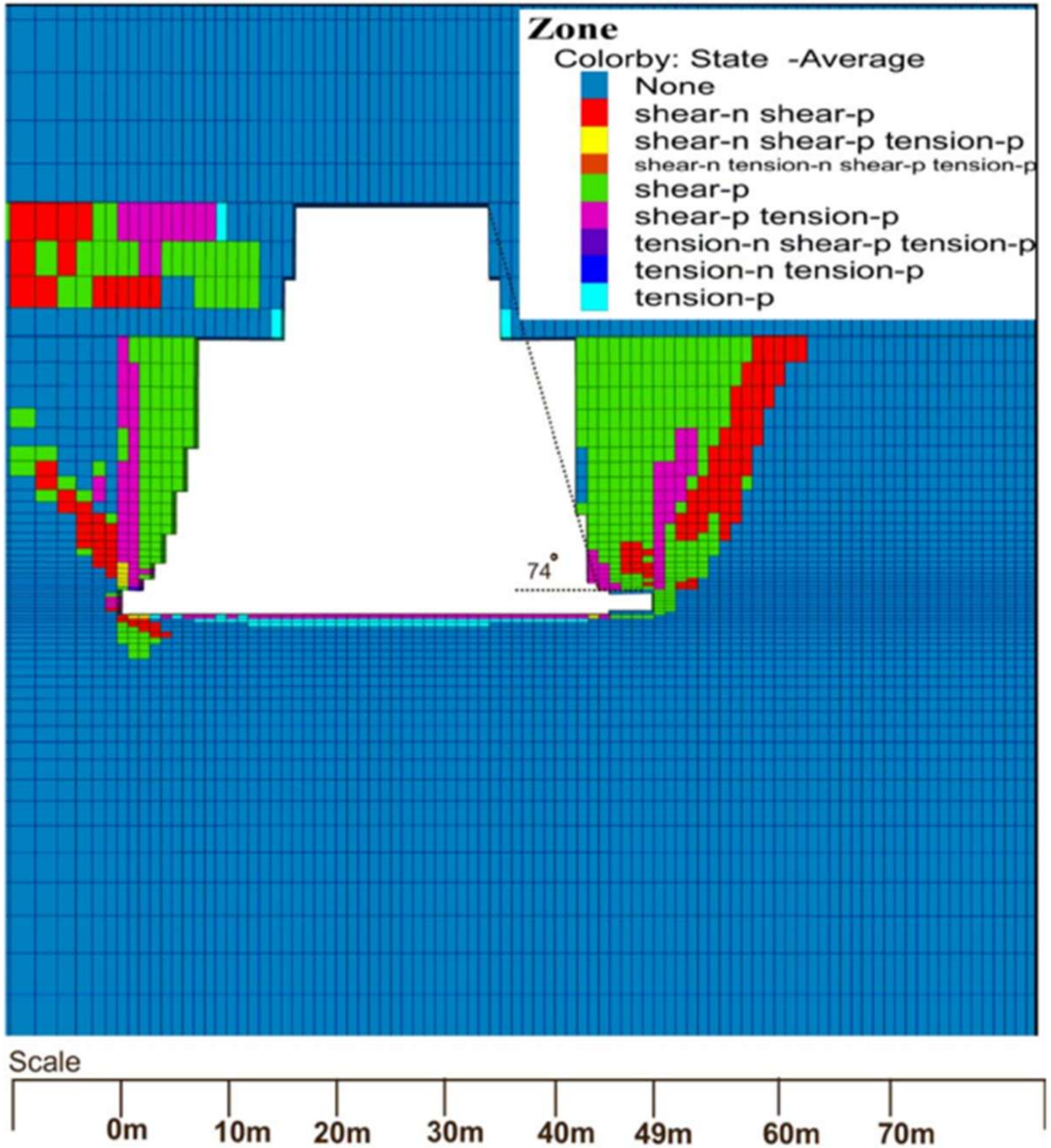


Zone
Colorby: Group Any
coal
floor
Immediate_roof
main_roof
overburden
Longwall face
Face support
Roller Boundary
0.85
1.10
1.15
Mining zone
2 times the mining zone
2 times the mining zone
0.90
Fixed Boundary
Z
Y
X




c Working 'B', 48m face advance

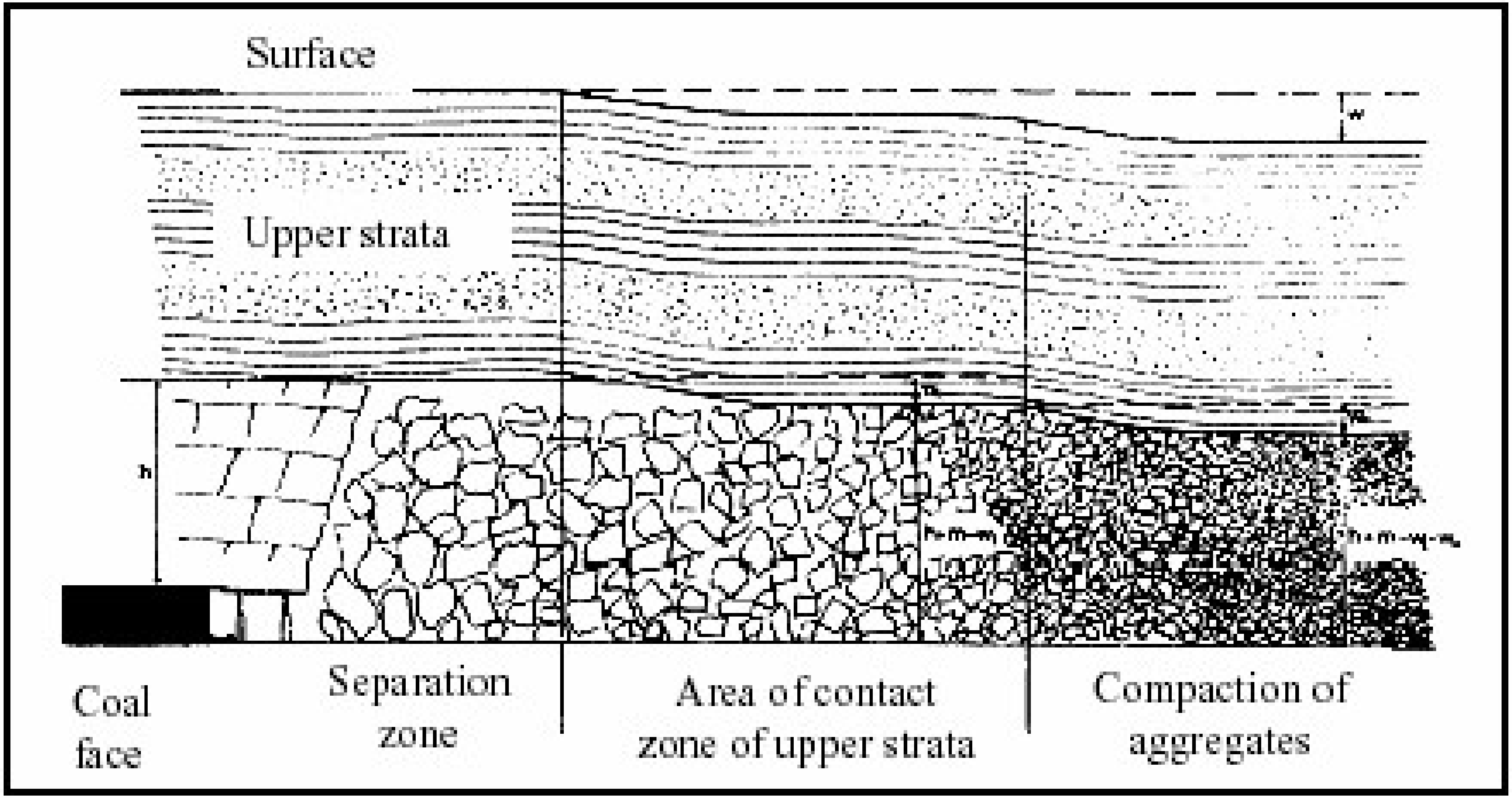
**First local fall**




**d** Working 'B', 49m face advance

**Main Fall**

# *Conceptual Sequence of Caving :*



## *Conceptual Sequence of Caving :*

- As the face advances, the span of unsupported roof in the goaf increases and the roof eventually caves in. This localized caving of immediate roof is known as **first local fall**. The caving starts well away from the face at about the middle of the exposed span.
- With the further face advance, the immediate roof continues to cave in inside the goaf and the caving line moves closer to the goaf edge of the support.
- The height of caving at this stage depends on the strata sequence of the immediate roof. If this roof consists of well stratified weak shales/sandstone, the height of caving may extend to the well defined bedding planes. In case of excessively hard sandstone strata, the roof may continue to hang in this phase without any sign of fracture or convergence.
- The immediate roof continues to cave in periodically without caving in of the main roof as the support moves forward with progressive advance of the face.

## *Conceptual Sequence of Caving :*

- When the face advances beyond a certain limit depending upon characteristics of the roof, the main roof overlying the immediate roof in the face area fails.
- The failure and subsequent caving of the main roof is known as **main fall**. It is accompanied by a substantial convergence of the roof in the face area.
- The front abutment stress at the face reaches the maximum value just before the occurrence of the main fall. It is considered that failure of strata starts occurring ahead of the face at the moment of the main fall.
- The main fall is most often accompanied by significant support closure, increase in load and spalling of coal from face. In some worst caving cases, air-blast, damage of supports and collapse of the face are also experienced, if the supports are not designed to cope with the intensive loading at the face.

# Assessment of Cavability and Support Resistance Estimation

- The estimation of cavability of overlying roof rocks is the first step for estimating the requirement of support resistance.
- Quantitatively, it is difficult to define cavability. But, a roof may be considered to be ideally cavable when the roof rocks cave in and fill the goaf as soon as the supports are withdrawn.
- The Longwall & Shortwall Mining Division of CIMFR has developed its own method for assessing the cavability of roof rocks.
- As has been observed, the caving behavior of different beds of overlying roof depends on the thickness, strength and massiveness of the bed. The following empirical relationship expresses the relation between caving index no. 'I' with different factors mentioned above:

$$I = \frac{\sigma_c L^n t^{0.5}}{5}$$

Where $\sigma_c$ = Compressive Strength, $Kg/cm^2$

L = Average length of core, cm

t = Thickness of bed, m

n = Constant depending upon the RQD of the bed.

# Assessment of Cavability and Support Resistance Estimation

| **Category** | **Nature of Caving** | **Rage of Cavability Index** | **Main fall span, $S_m$, m** | **Weighting Dynamism** |
|---|---|---|---|---|
| Category-I | Easily cavable | $I < 2000$ | $S_m < 35$ | --- |
| Category-II | Moderately cavable | $2000 \leq I < 5000$ | $35 \leq S_m < 55$ | --- |
| Category-III | Cavable with difficulty | $5000 \leq I < 10000$ | $55 \leq S_m < 80$ | May/may not be |
| Category-IV | Cavable with substantial difficulty | $10000 \leq I < 14000$ | $80 \leq S_m < 95$ | May/may not be |
| Category-V | Cavable with extreme difficulty | $\geq 14000$ | $S_m \geq 95$ | Dynamic |

# Assessment of Cavability and Support Resistance Estimation

## ***Estimation of Support Resistance:***

1. *Caving properties of the strong bed, $I_{max}$*

- This is indicated by 'Caving Index Number' 'I' of the strong bed. This index depends on the thickness of strong beds, its strength properties and massiveness.
- Higher the index, larger is the span at which first weight (and also subsequent periodic weights) occurs and more severe is the weighting and worse is the resultant roof conditions.
- Caving index number of all the beds occurring within twelve to fifteen times the height of extraction would have to be analyzed to identify the bed, which will have predominant influence in causing the weighting.

# Assessment of Cavability and Support Resistance Estimation

2. *Thickness of cavable beds in between strong bed and the coal seams in terms of height of extraction:*

- If the thickness of cavable bed is adequate to fill the goaf completely, then a part of the strain energy released during breakage of the strong bed would be absorbed by the caved material in the goaf and severity of first weight and periodic weights would be lessened.
- But if the thickness of cavable bed is inadequate to fill the goaf then the first weight and the periodic weights would be dynamic and sever in nature unless artificial means of caving is practiced.

*3. Resistance offered by the support system*

- The increase in support resistance has a tendency to decrease the severity of first weight and periodic weights improving the resulting strata control condition.

# Assessment of Cavability and Support Resistance Estimation

$$C_m = \left( \frac{A}{P} + 9.6 h_e \right) + \left( \frac{k I_{max}}{k_1 + 1.5} - 23 \right)$$

Where $A$ = Constant depending upon the category of the roof

$C_m$ = Maximum convergence during first weight & periodic weights, mm/m

$P$ = Support resistance, t/m$^2$ ***(MLD)***

$I_{max}$ = Cavability Index of strong bed,

$K_1$ = A factor depending on the thickness 't' of cavable bed in between strong bed and the coal seam in terms of height of extraction.

$h_e$ = height of extraction, m

# Assessment of Cavability and Support Resistance Estimation

- A is an empirical constant, depending on category of roof (Refer Table 1); its value is taken as 1440 for roof category I and II, 1700 for roof category III and IV, and 1900 for roof category V
- $K_1$ is a factor depending on the ratio (r) of thickness of cavable bed in between the coal seam and the strong bed to the height of extraction, its value is taken as 2 for $r \leq 2$, 3 for $2 < r \leq 4$, 5 for $4 < r \leq 8$ and 10 for $r > 8$
- K is a constant, 0.025

# Assessment of Cavability and Support Resistance Estimation

- With the help of the suggested relationship, it is possible to project maximum convergence at a proposed longwall face (i.e. under a particular strata condition) with different values of mean load density.

- As the rates of convergence have already been correlated with degradation of roof on the basis of measurements conducted over a few hundreds of cycles it is possible to project likely roof condition with a particular value of mean load density.

- A value of mean load density would be chosen which would give safe roof condition during the occasions of weighting.

# Assessment of Cavability and Support Resistance Estimation

| Maximum Convergence | Expected roof condition |
|---|---|
| Maximum convergence 60 mm/m | Convergence within permissible limit |
| Above 60mm/m to 100mm/m | Minor roof fracturing increasing with the value of convergence. |
| Above 100 up to 160 mm/m | Significant roof fracturing and roof degradation. Seriousness increasing with increase in convergence. |
| Above 160 mm/m | Rock fall zone. |

- For ***Indian geo-mining conditions*** as shown in the above table, for ***limited and acceptable degradation of the roof*** a ***maximum convergence*** at the face is taken as ***60 mm/m***. Effective Support Resistance for different height of extraction is calculated based on the above relationship

## Problem 2

Refer Excel Sheet

# Assessment of Cavability and Support Resistance Estimation

**Rated support resistance**

However the Rated Support Resistance should take into account the following deficiencies during the actual operation –

I. Leakage in leg circuit: 10%

II. Setting load deficiencies: 10%

III. Miscellaneous (deviation from normal span, Premature bleeding of leg circuits, etc.) :10%

Hence the overall deficiency to be taken into account is 30% and thus, the Rated Support Resistance is calculated from the Effective Support Resistance with a safety factor of 1.3.

# Assessment of Cavability and Support Resistance Estimation

## SUPPORT CAPACITY ESTIMATION

$$\text{Support Capacity (in T)} = P_r \; W_c \, (L_c + W_d + 0.3)$$

where, $P_r$ is the rated support resistance, $T/m^2$

$W_c$ is the width of the canopy, m

$L_c$ is the length of the canopy, m

$W_d$ is the web depth of shearer, m

*Assume,* $Wc = 1.75$m, $Lc = 4.79$m and $W_d = 0.85$ m

# Problem 3:

| Bed No | Avg Rock Type | Bed thickness, m | RQD, % | Avg. Core Length, cm | Density, gm/cc | CS, ksc | TS, ksc | CI | RMR |
|---|---|---|---|---|---|---|---|---|---|
| | coal | 3.50 | 40 | 4.37 | -- | 281 | -- | 0 | 43 |
| 1 | shcoal | 3.70 | 52 | 8.3 | -- | 215 | -- | 686 | 42 |
| 2 | mcgsst | 2.79 | 82 | 18.26 | 2059 | 209 | 27 | 2276 | 50 |
| 3 | cgsst | 2.95 | 88 | 20.27 | 2041 | 163 | 16 | 2070 | 50 |
| 4 | cvcgsst | 3.05 | 56 | 9.56 | 2031 | 128 | 9 | 948 | 33 |
| 5 | mcgsst | 17.00 | 89 | 20.55 | 2055 | 192 | 24 | 6326 | 53 |
| 6 | mcgsst | 1.80 | 68 | 13.54 | 2073 | 185 | 27 | 1593 | 45 |
| 7 | mgsst | 2.74 | 88 | 20.32 | 2071 | 248 | 34 | 3041 | 57 |
| 8 | shale | 0.54 | 98 | 23.56 | -- | 40 | -- | 139 | 46 |
| 9 | fgsst | 1.08 | 92 | 21.42 | 2240 | 312 | 44 | 2564 | 69 |
| 10 | carb_coal_clay | 3.79 | 96 | 22.87 | -- | 173 | -- | 1538 | 43 |
| 11 | fmgsst | 1.68 | 43 | 5.51 | 2063 | 277 | 38 | 1293 | 38 |
| 12 | sandy_sh | 2.09 | 99 | 23.86 | -- | 192 | -- | 1328 | 45 |
| 13 | cgsst | 5.28 | 95 | 22.61 | 2036 | 177 | 18 | 3426 | 62 |

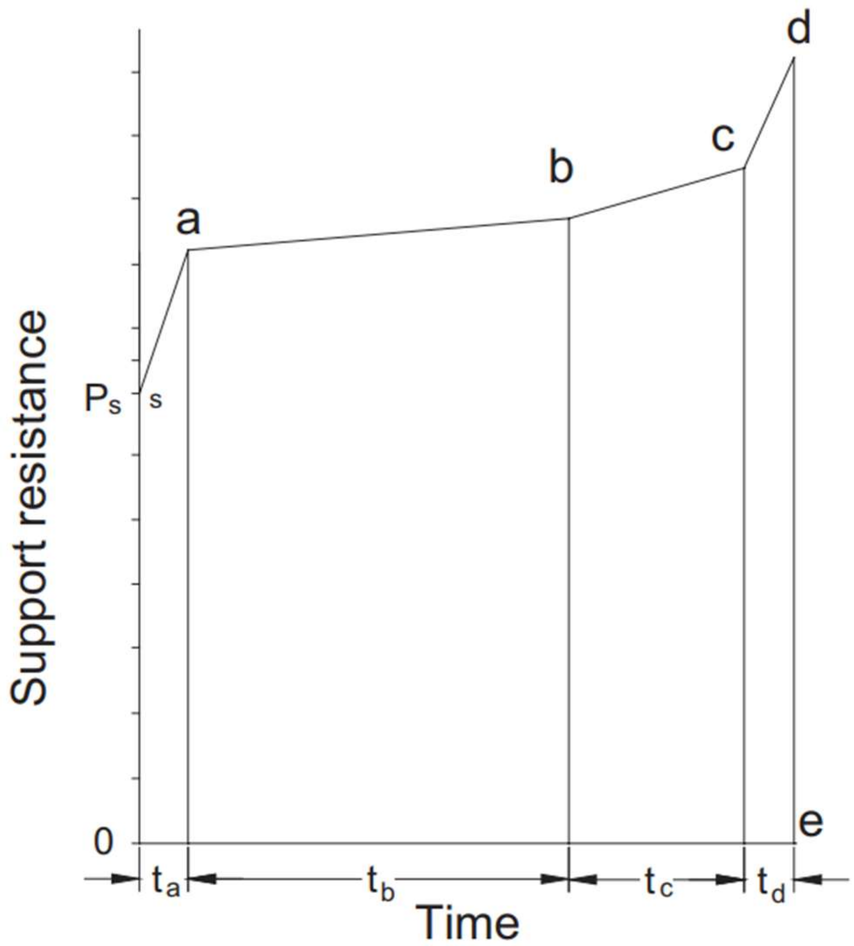
# Assessment of Cavability and Support Resistance Estimation

- The approach also estimates the span of main fall (Sa) and periodic caving (Sp)

$$S_a = 0.72 I^{0.51}$$

$$S_p = 3.05 + 0.25 S_a$$

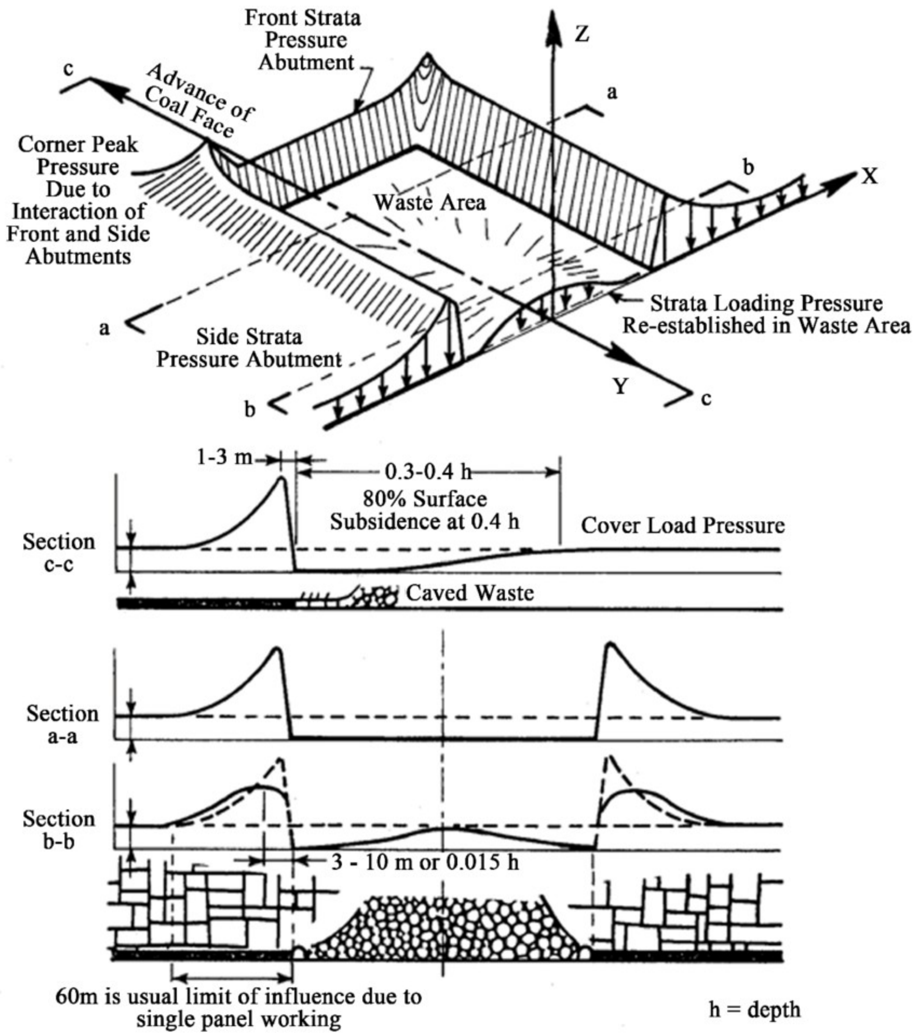
# Supporting Cycle

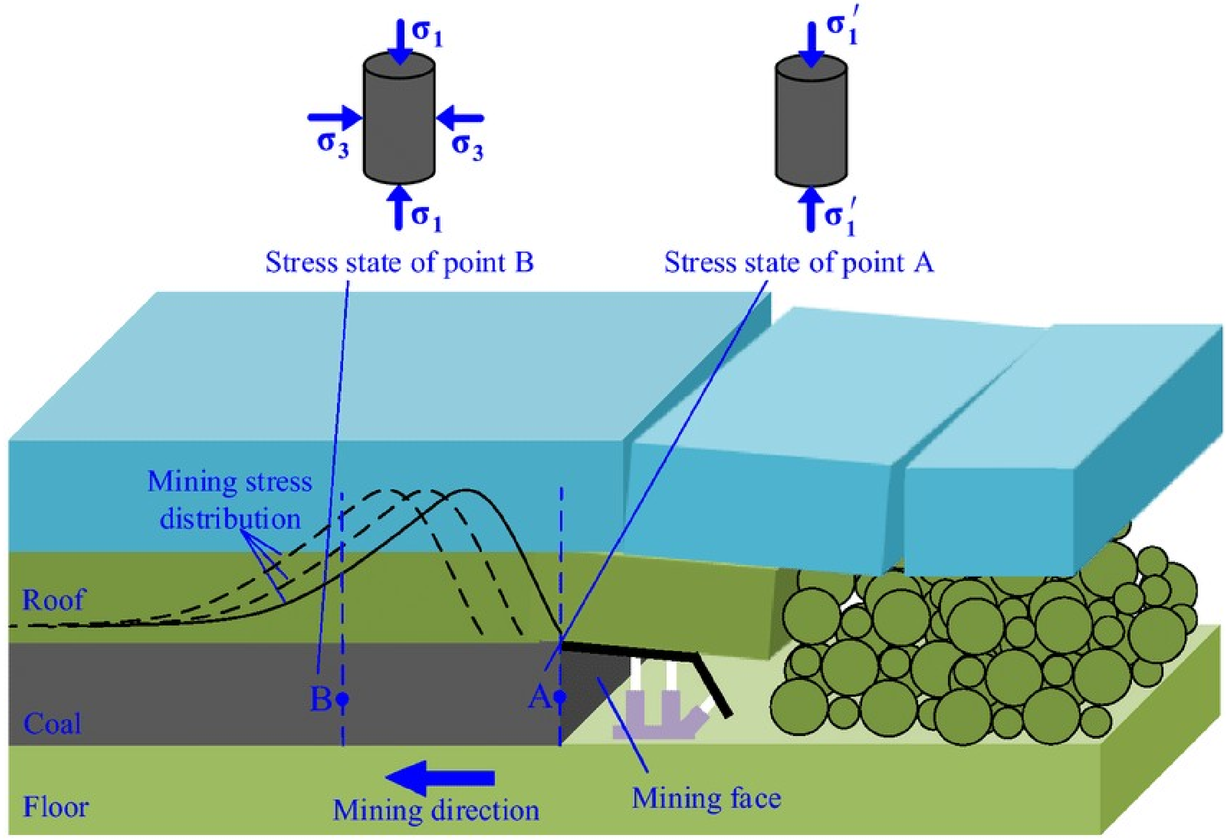



- After a shield is set against the roof and floor, its load (or actually resistance to roof convergence and/or floor heave) does not remain constant at the setting load level. Rather it varies from period to period as a result of interaction among roof strata, shield, and floor strata.

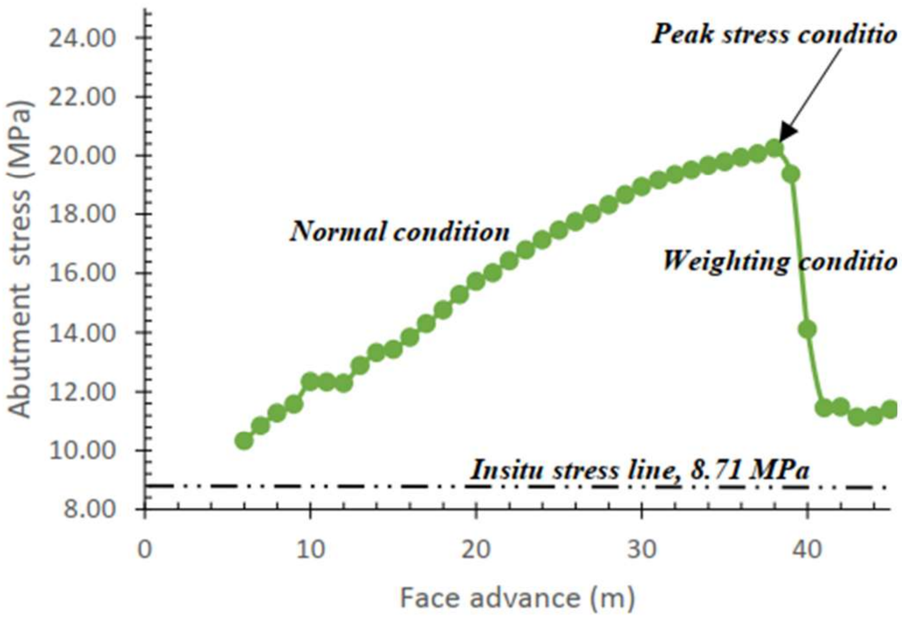
# Ground Control Considerations

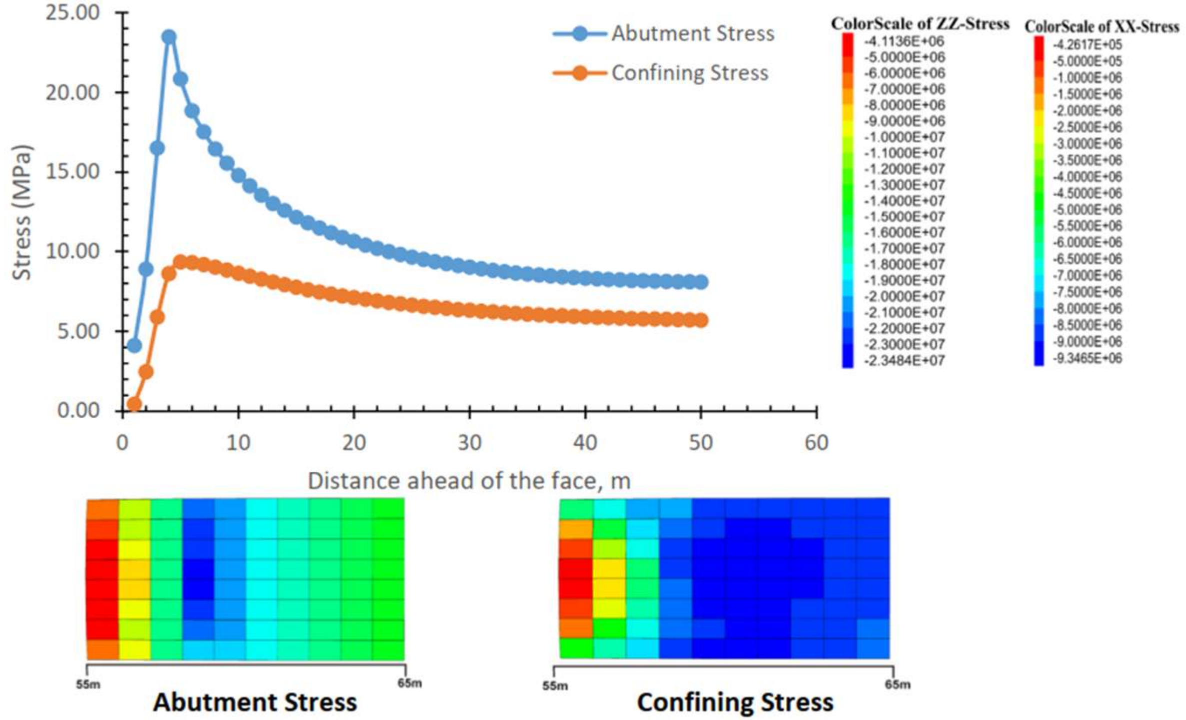
## Abutment Stress



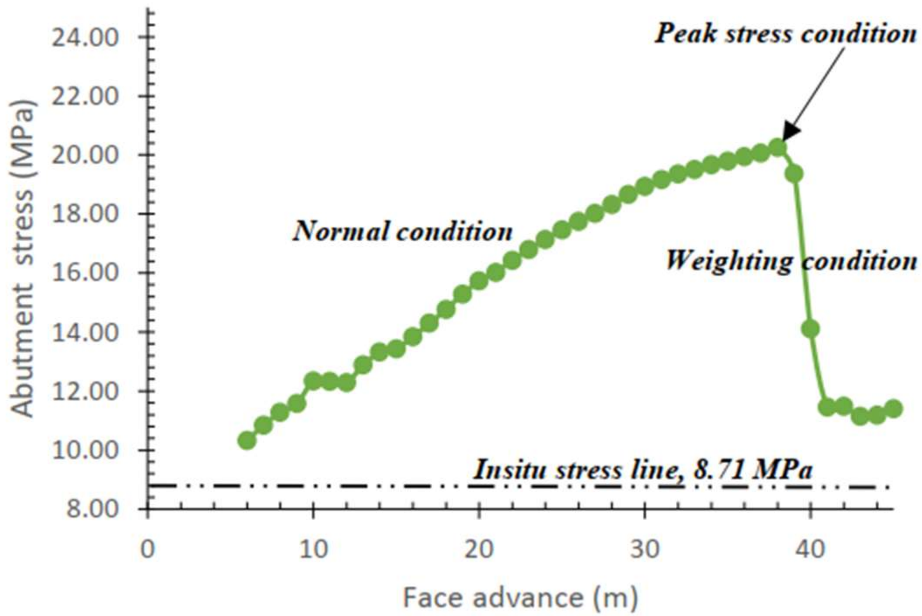



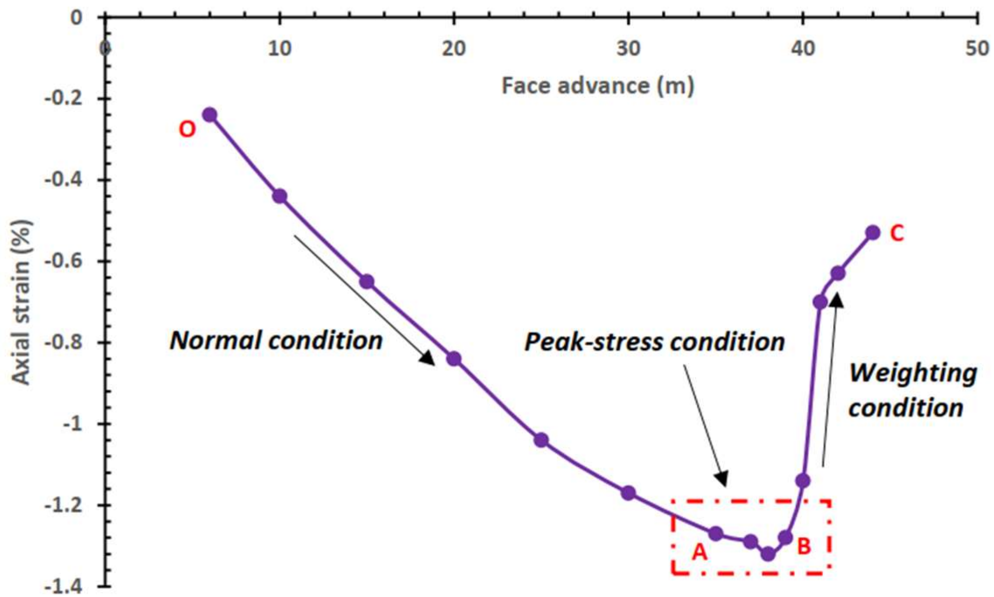
# Abutment and Confining Stress



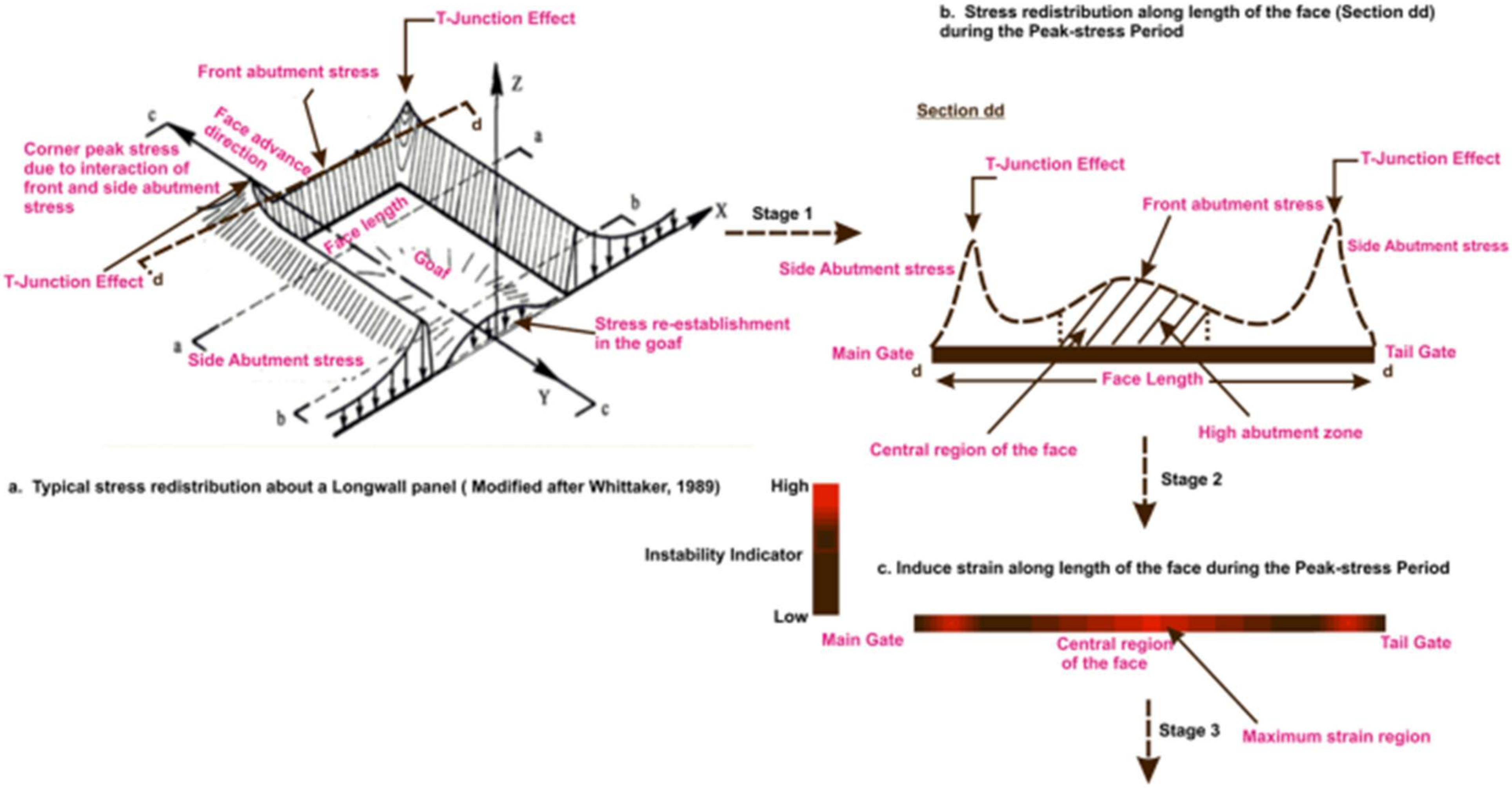



# Abutment Stress and Axial Strain



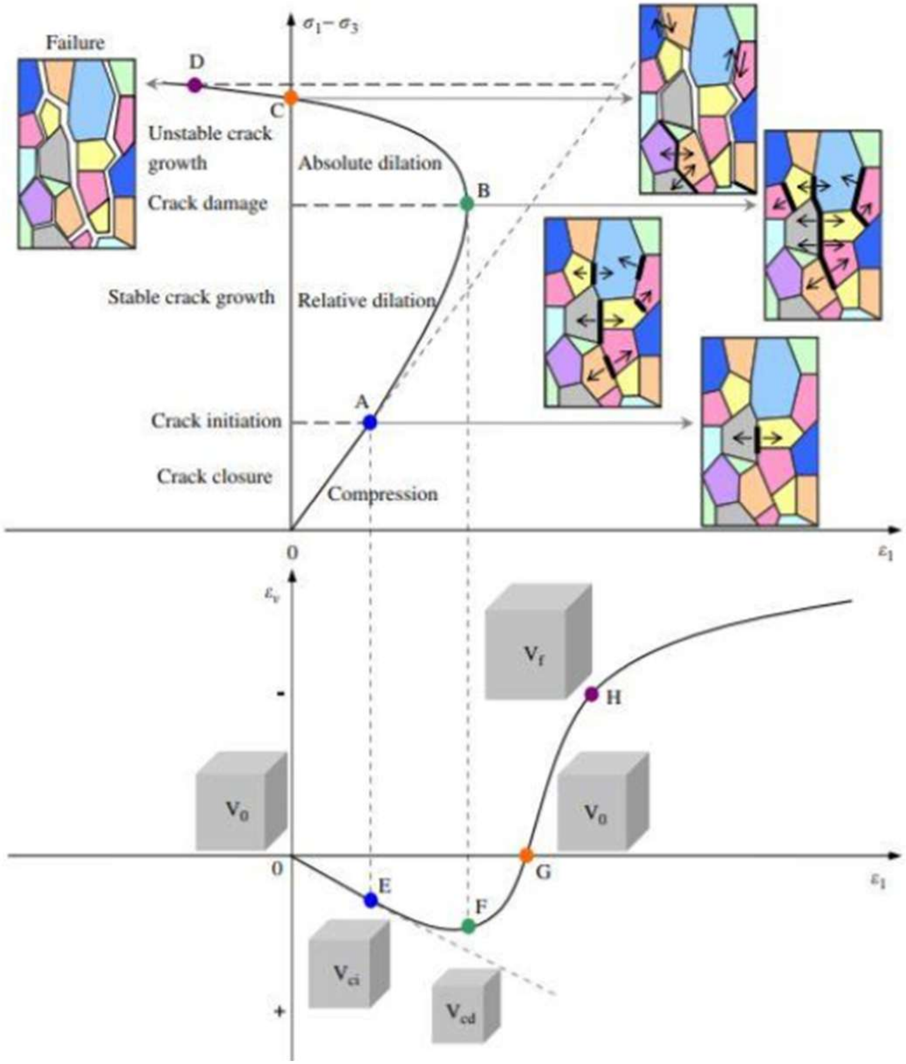
# Abutment Stress and Axial Strain along Face Length

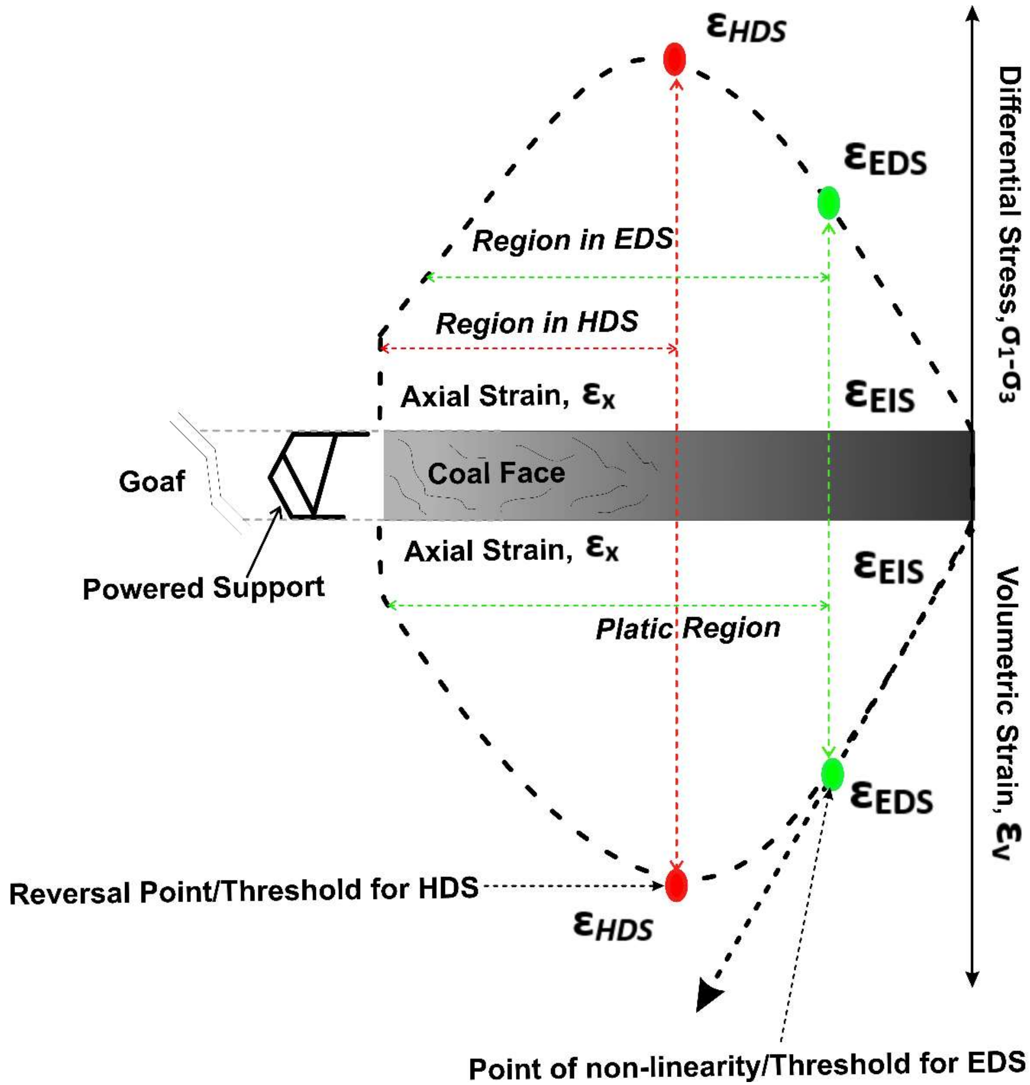


# Abutment Stress

- The redistribution pattern of front abutment stress is significant for quantification of damage and instability along the length of a face during its progressive advance.
- The operational dynamics of longwall operation is a function of peak abutment stress, which varies with the progressive mining cycle.
- The magnitude of the front abutment is the maximum in the central part of a face, and this zone of peak abutment varies linearly with face length
- For longer face length, the area of the maximum abutment zone at the centre of the face could be very high experiencing increased loading in comparison to the smaller face length.
- The state of loading can be catastrophic in deeper longwall operations due to higher stress environment.
- As the abutment zone also varies for different face length/cover-depth, supercritical longwall faces experience higher abutment along the face compared to the critical and sub-critical face operations.

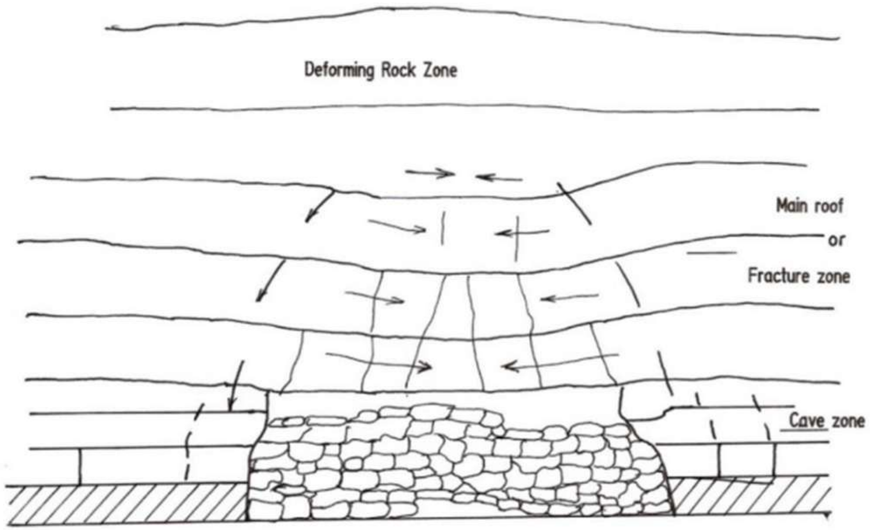
# Mechanics of Face Damage

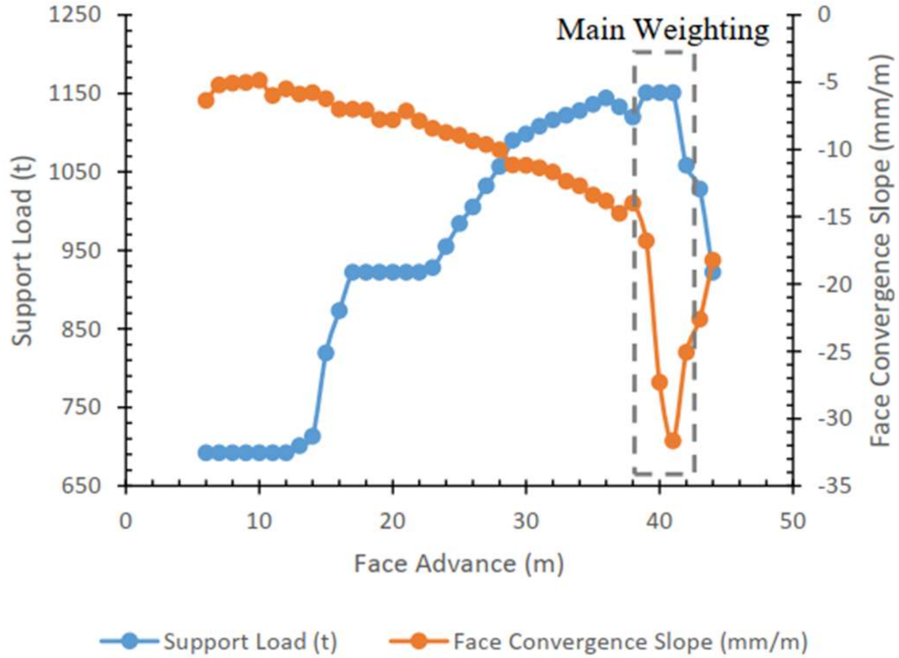


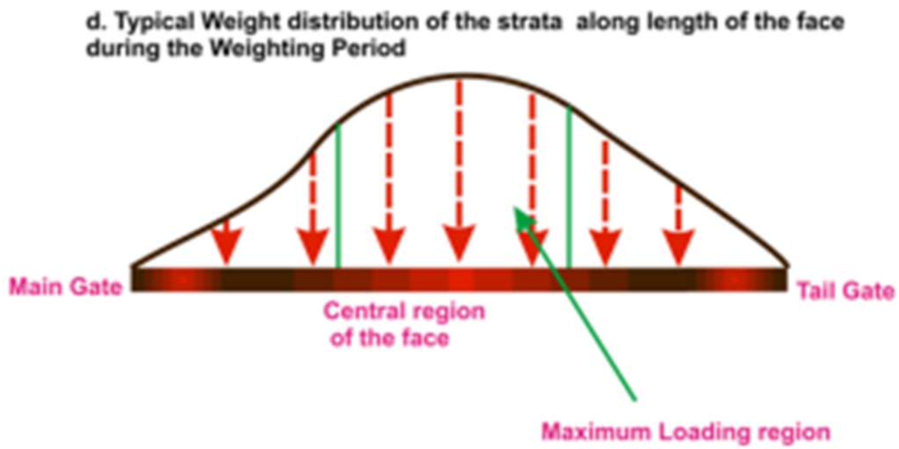
**Stress-Induced Face damage**

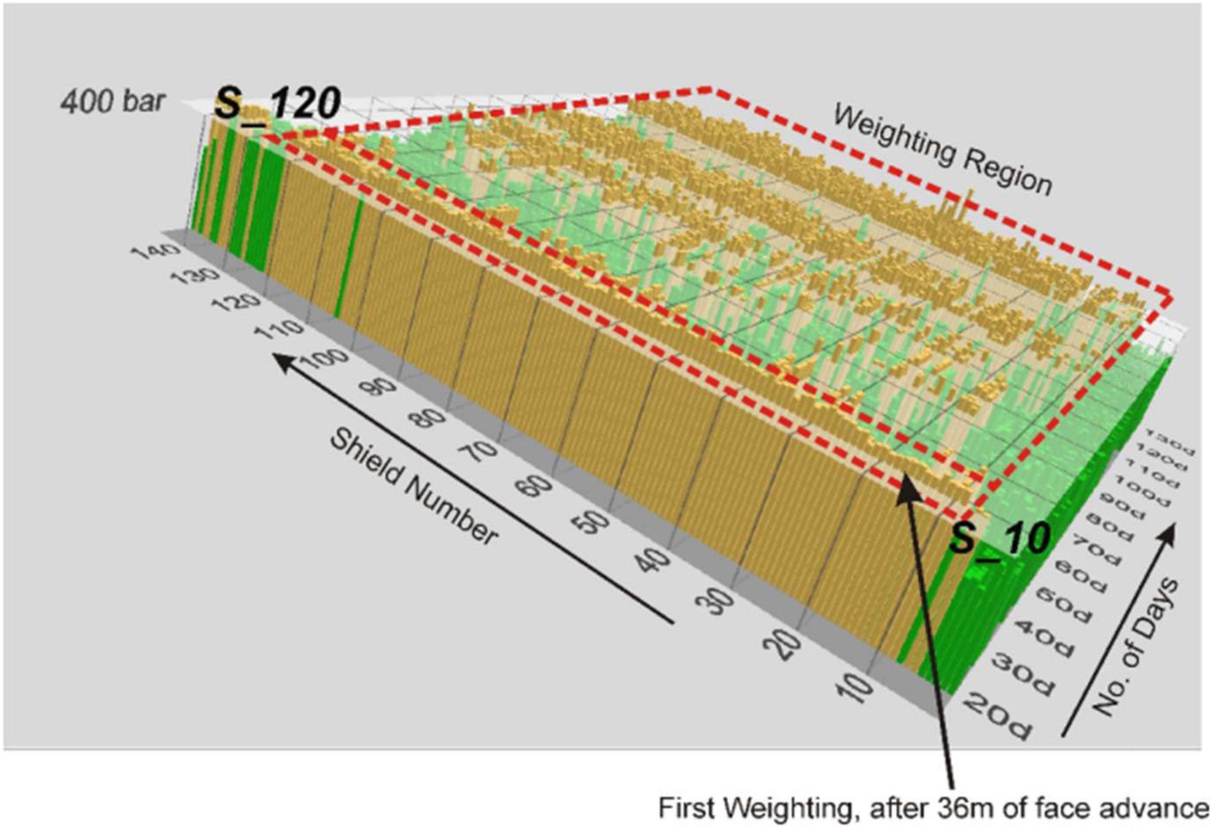
# Roof Weighting Mechanism



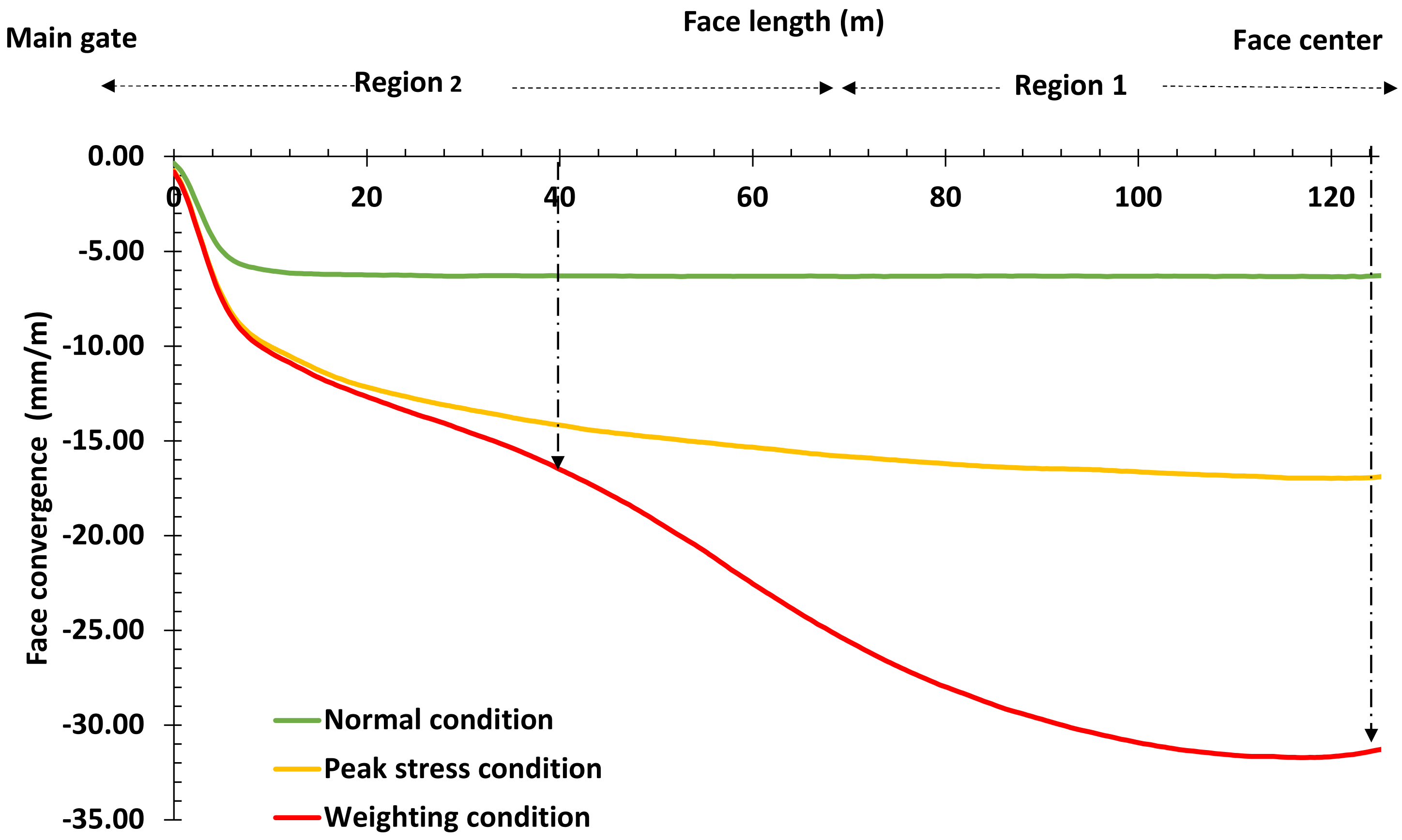



# Roof Weighting Mechanism

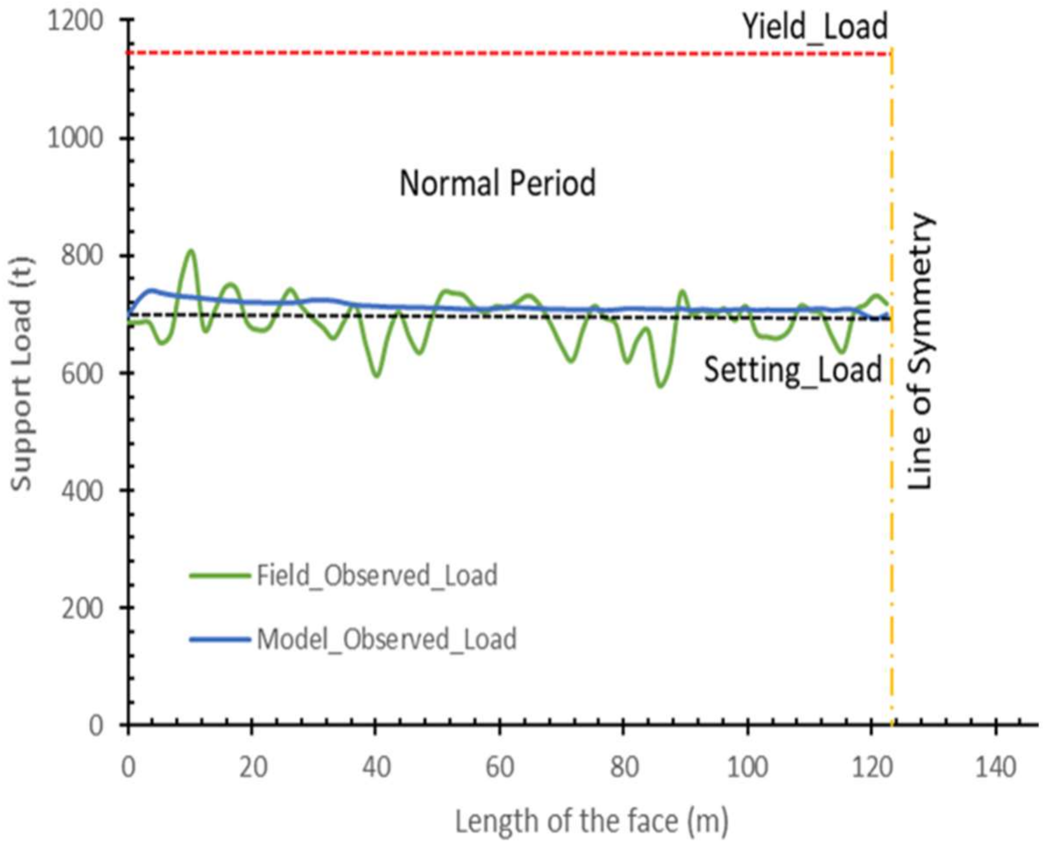


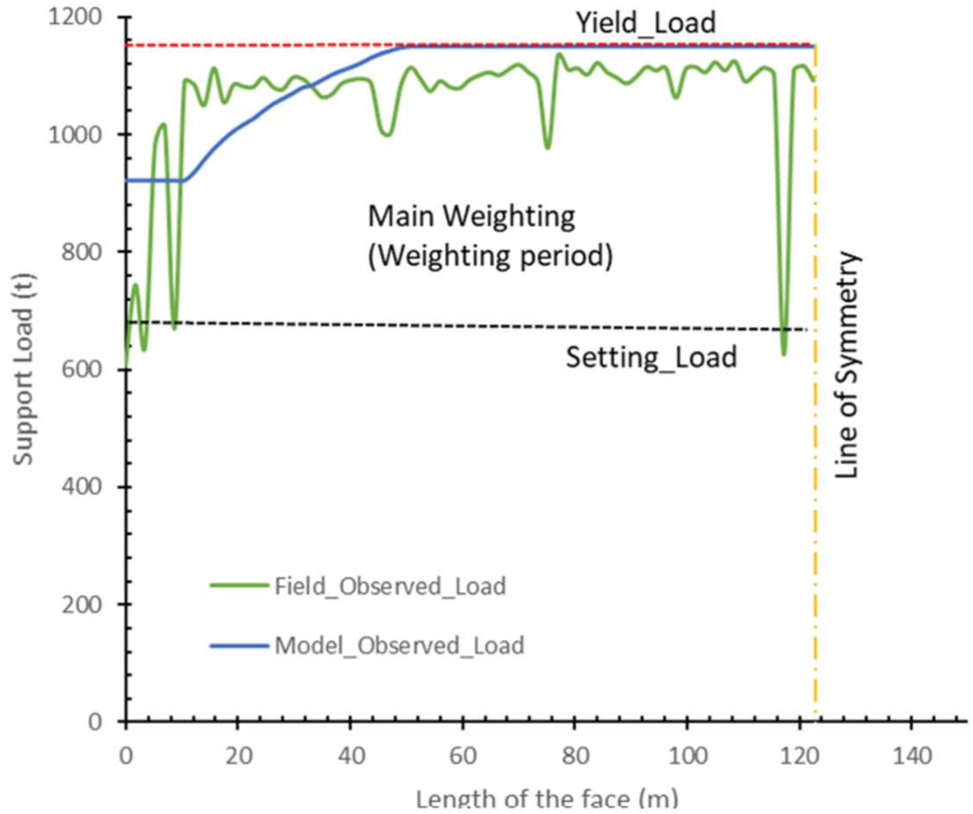



# Face Convergence

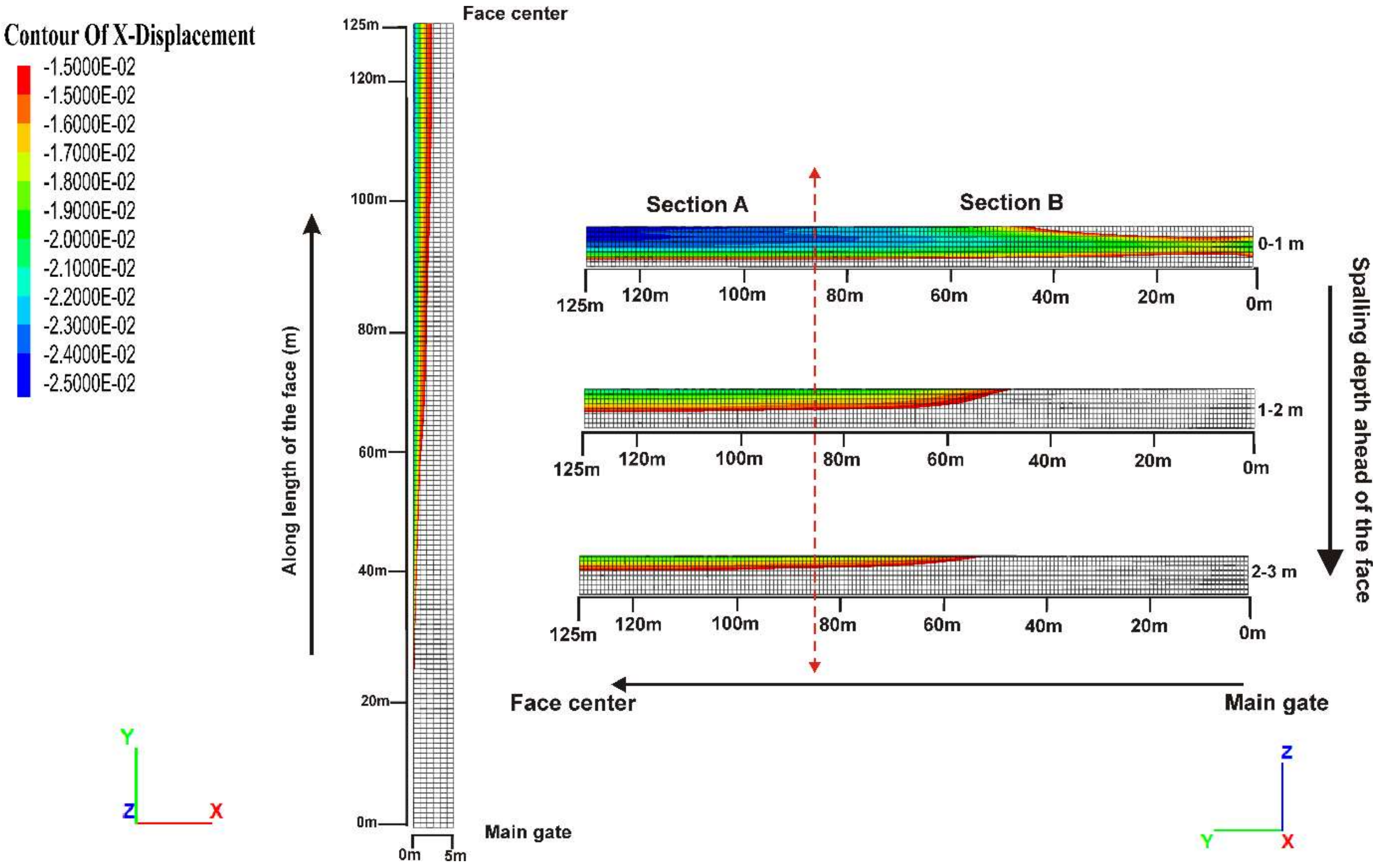


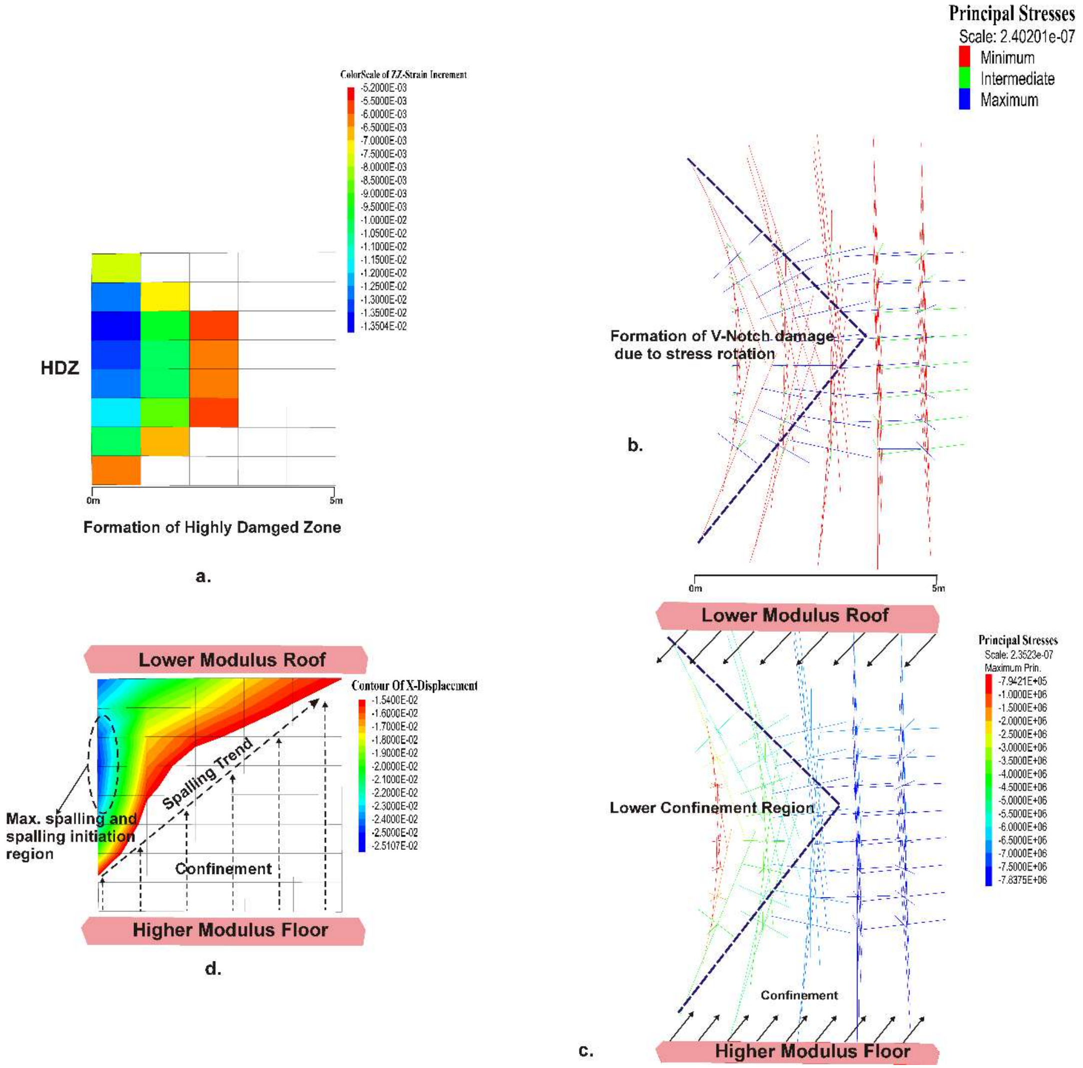
# Support Loading on PRS



# Face Spalling



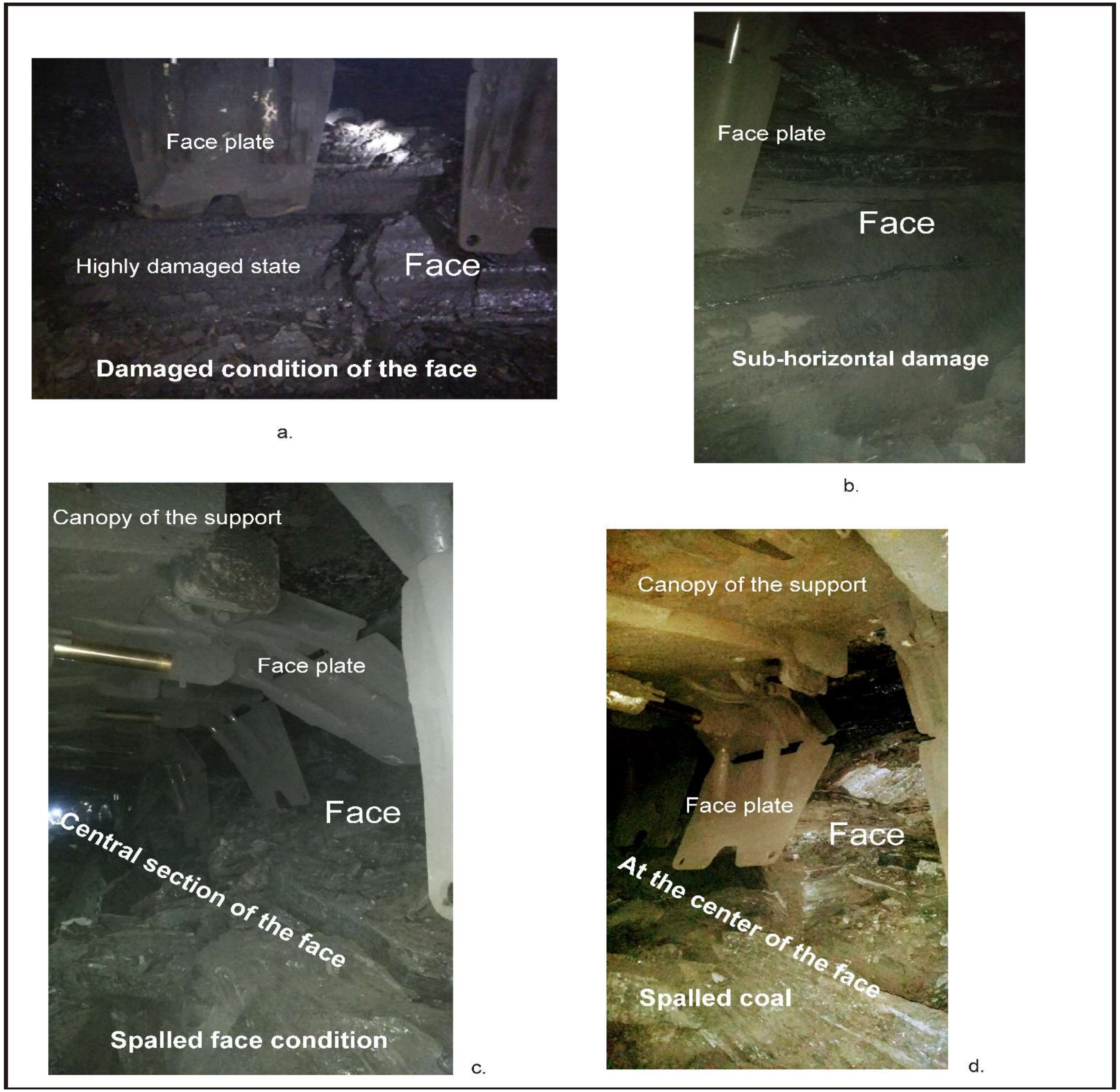



***Stress-Induced Face Damage to Weighting-Driven Face Spalling follow a Cyclic Process***


Face Spalling


Face Spalling


Face Spalling

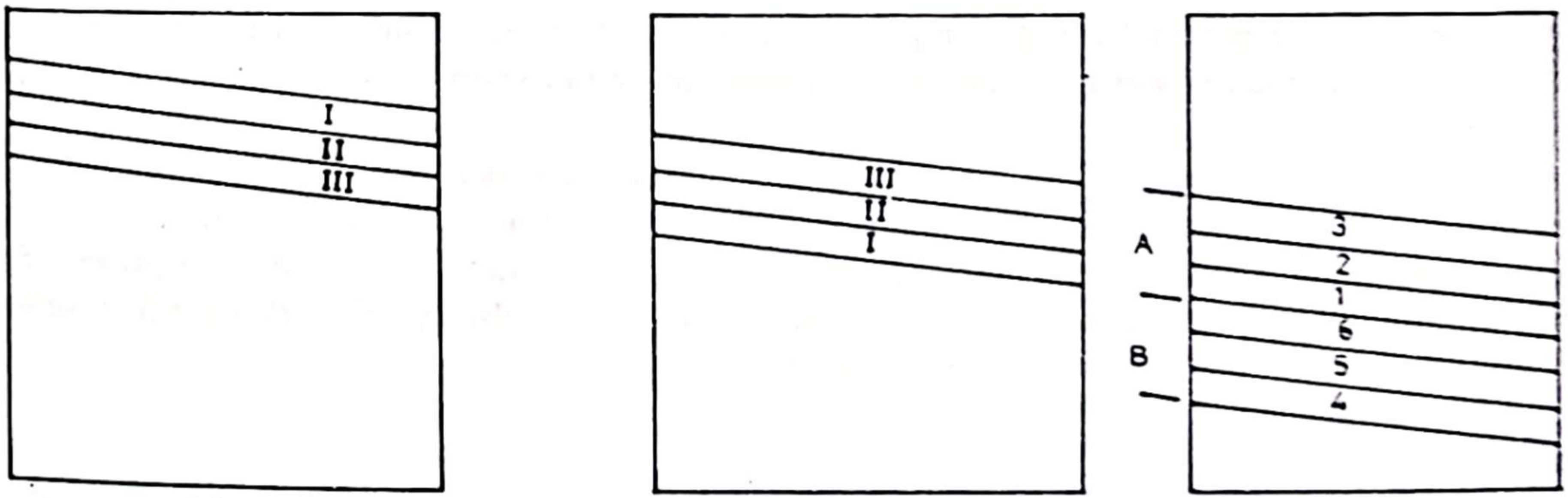

Face plate
Highly damaged state
Face
Damaged condition of the face
a.
Face plate
Face
Sub-horizontal damage
b.
Canopy of the support
Face plate
Face
Central section of the face
Spalled face condition
c.
Canopy of the support
Face plate
Face
At the center of the face
Spalled coal
d.

**Summary on Face Spalling Observations**

- The study showed that working deep coal seams with wider face length and higher extraction height could create considerable potential for face instability due to the higher volume of face spalling.
- The study also showed that working under soft and hard strata formation could pose a significant strata control challenge, where the amount of face spalling was higher than the moderate strata condition.
- working under a stronger roof with a larger face dimension could worsen the strata control issue associated with higher intensity of strata weighting and face spalling. It was also revealed that the increased thickness of the main roof could elevate the face spalling at the coal wall, which reduced the thickness of the main roof.

# *Thick Seam Mining Methods*

## Slice Mining




(a) DESCENDING

(b) ASCENDING

(c) ASCENDING-DESCENDING

## *Thick Seam Mining Methods*

- Ascending Vs. Descending Slicing
- Thickness of Slices
- Disadvantages of too-thick slices
- Factors affecting the division of a seam into slices

## Main Slicing Methods

### *Inclined Slicing*

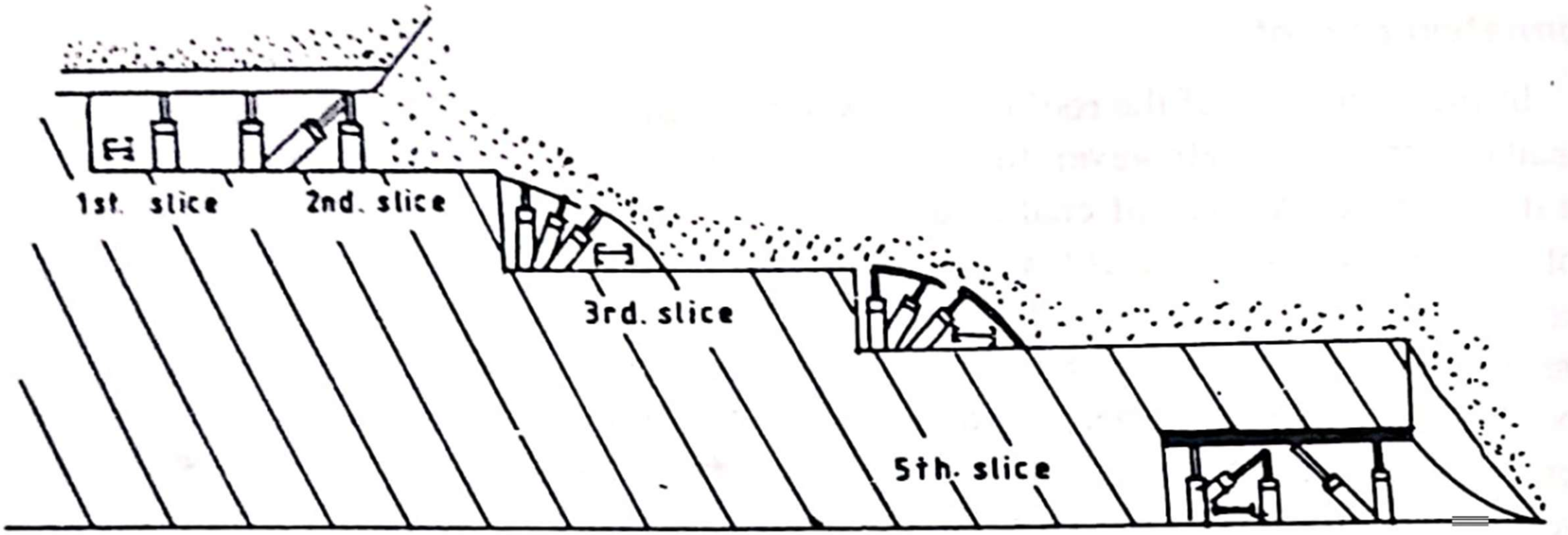
- Applicability
- Inclined Slicing in Descending Order with Caving
- Inclined Slicing in Ascending Order with Stowing

### *Horizontal Slicing*

- Applicability
- Horizontal Slicing in Ascending Order with Stowing
- Horizontal Slicing in Descending Order with Caving

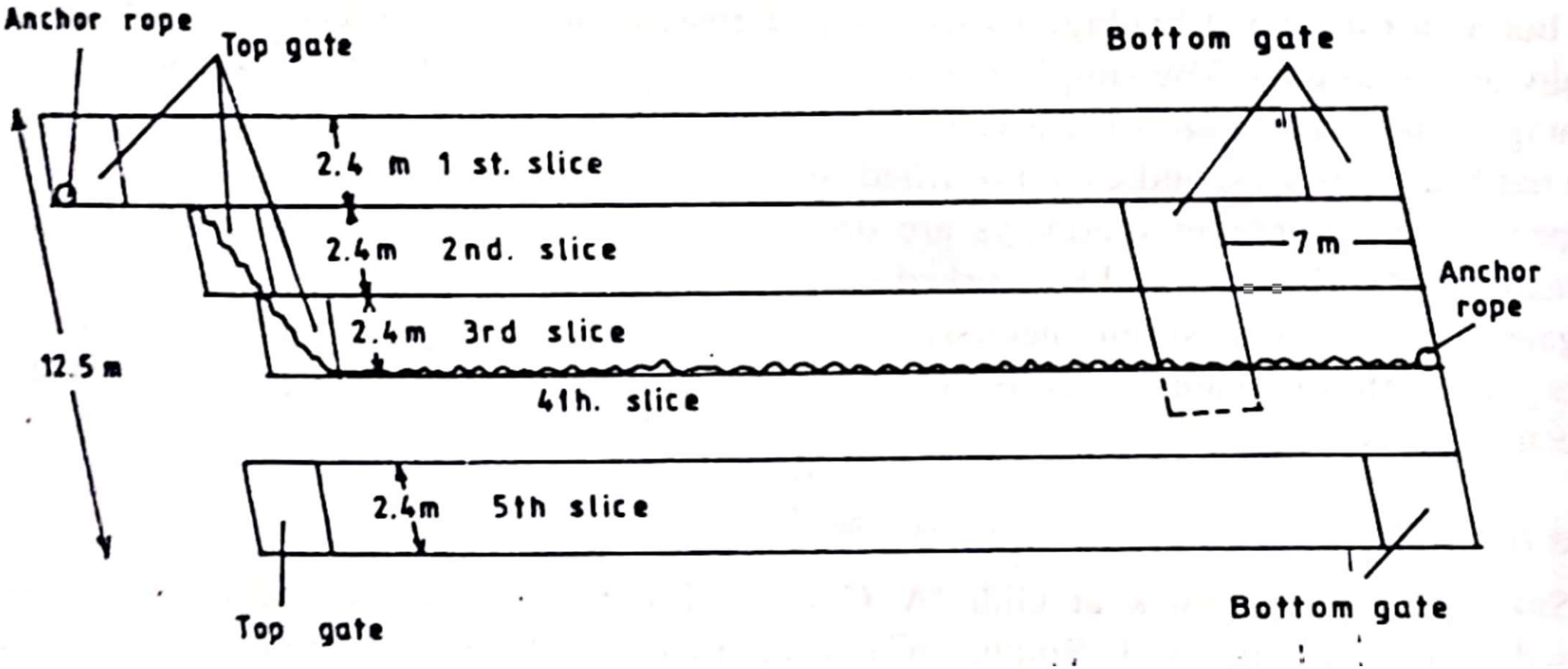
# Thick Seam Mining Methods

***Inclined Slicing*** in ***Descending Order with Caving***

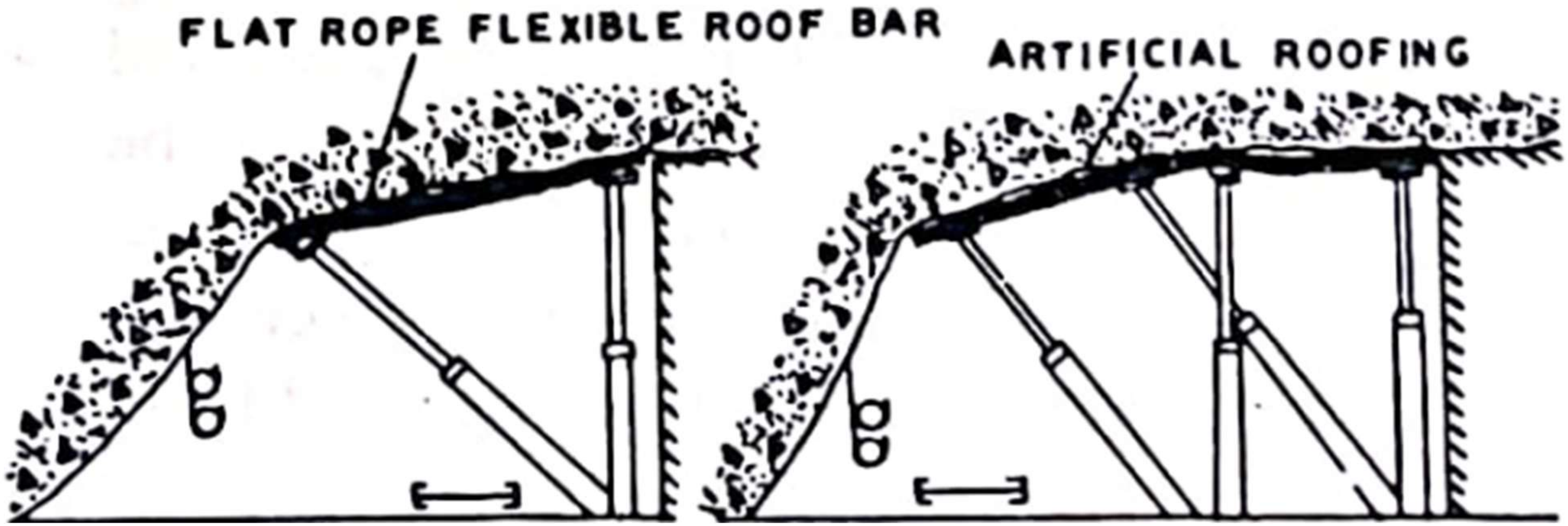


# *Thick Seam Mining Methods*

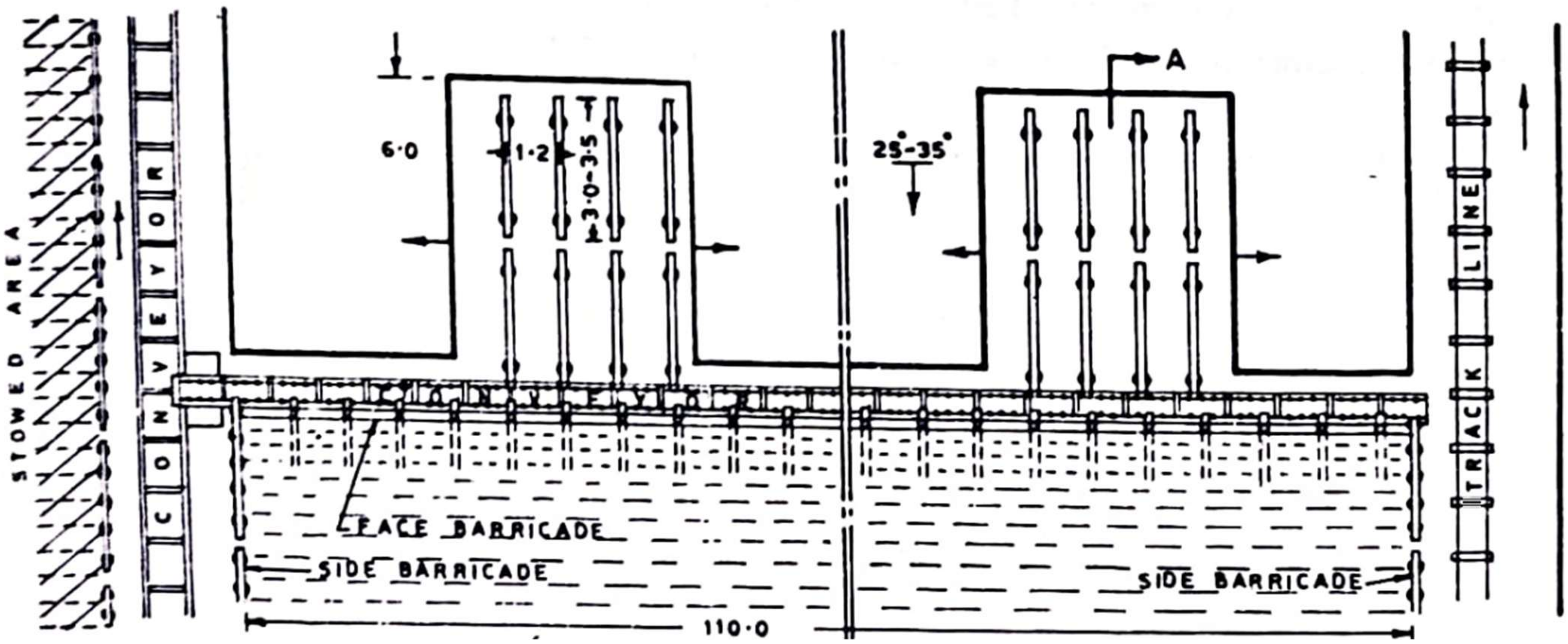
***Inclined Slicing*** in ***Descending Order with Caving***



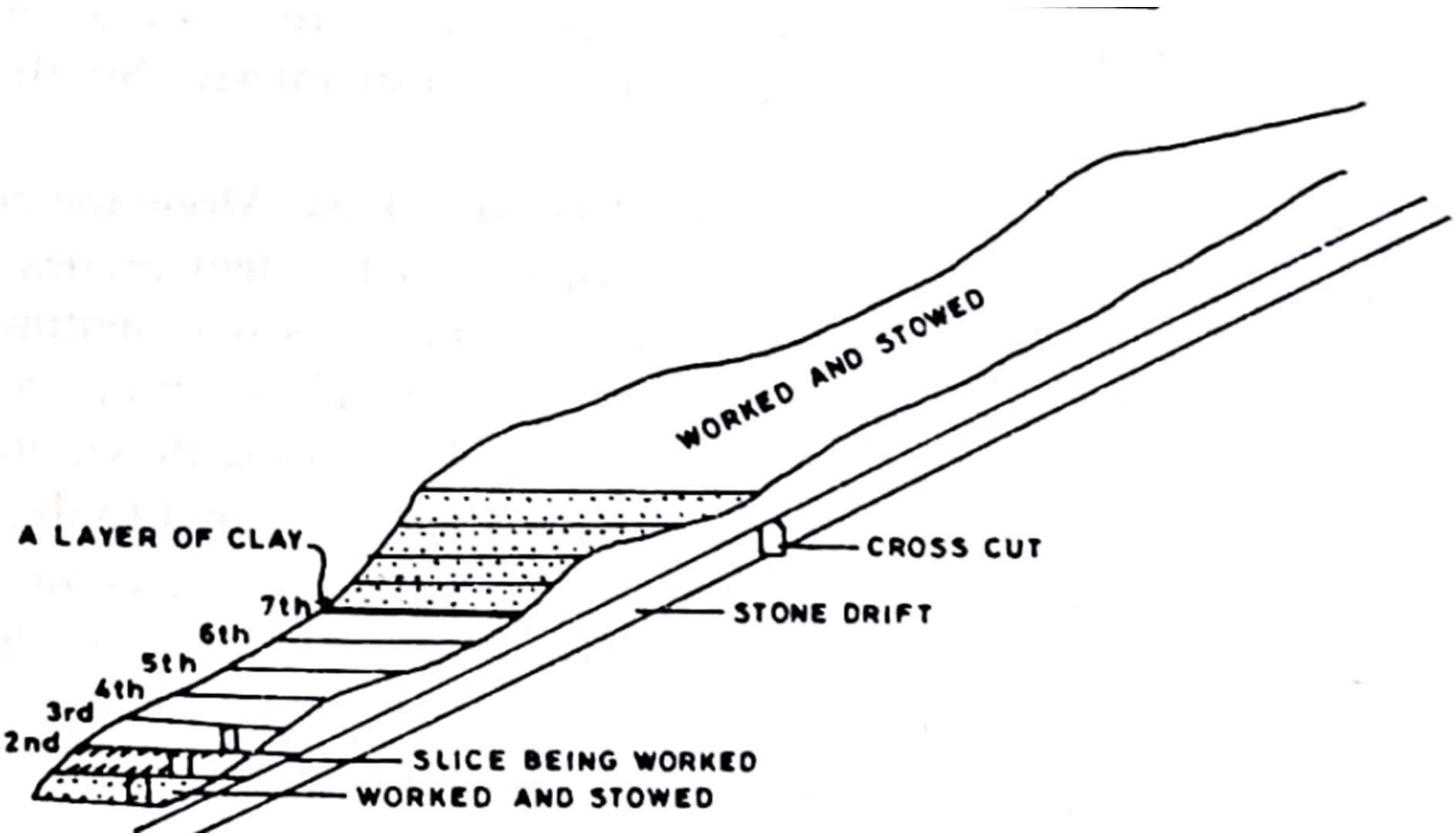
***Inclined Slicing*** in ***Descending Order with Caving***



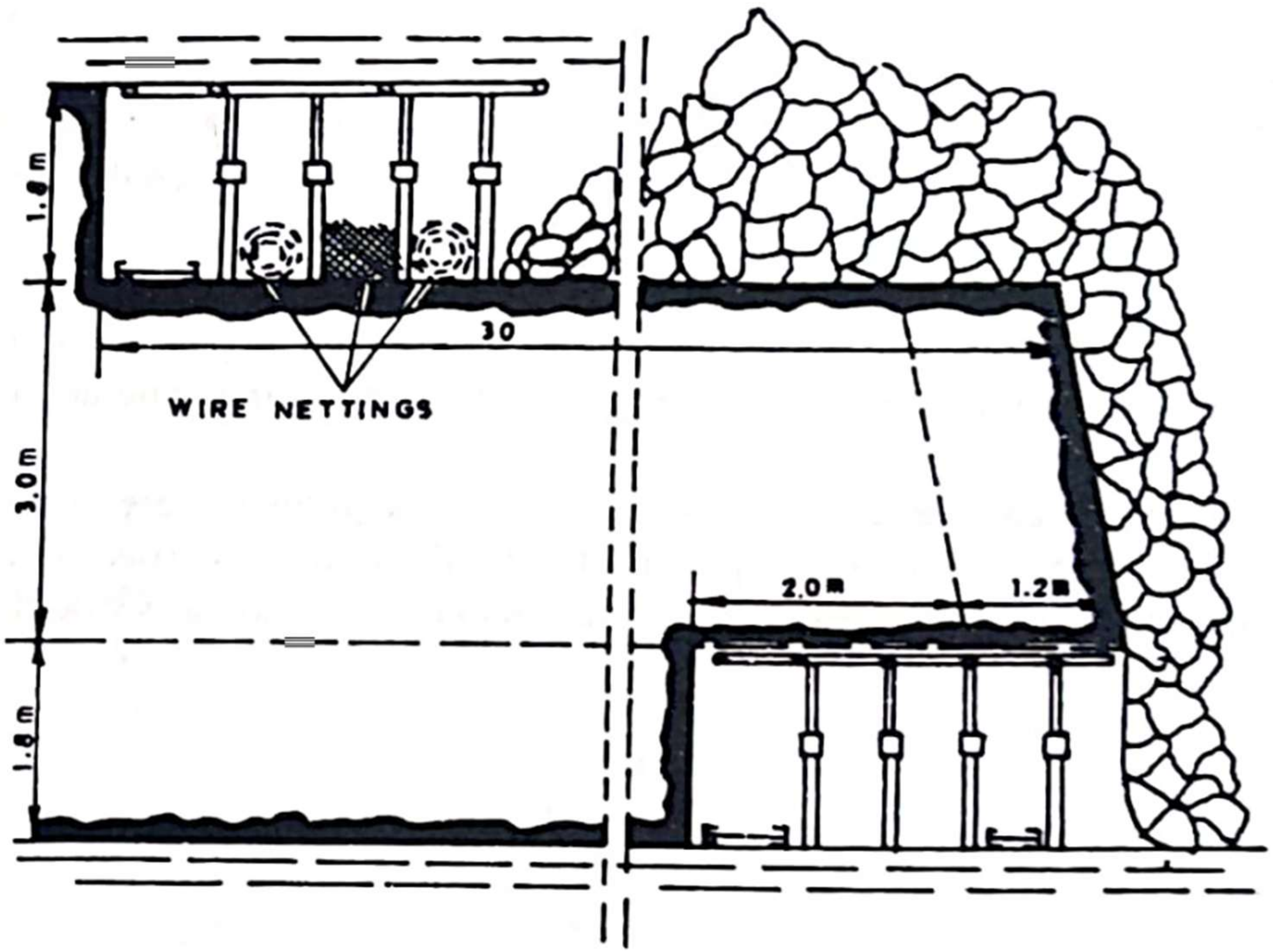
## ***Inclined Slicing*** in ***Ascending Order with Stowing***



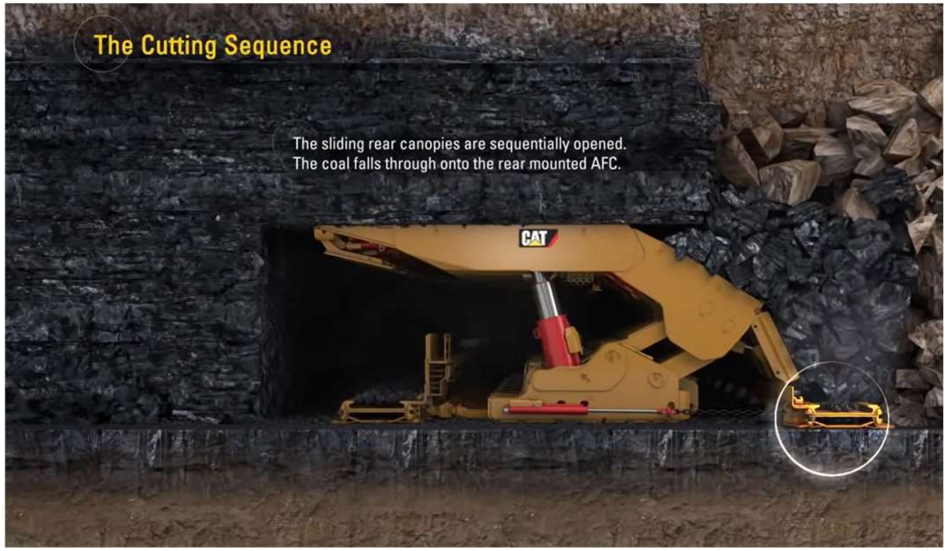
# Horizontal Slicing

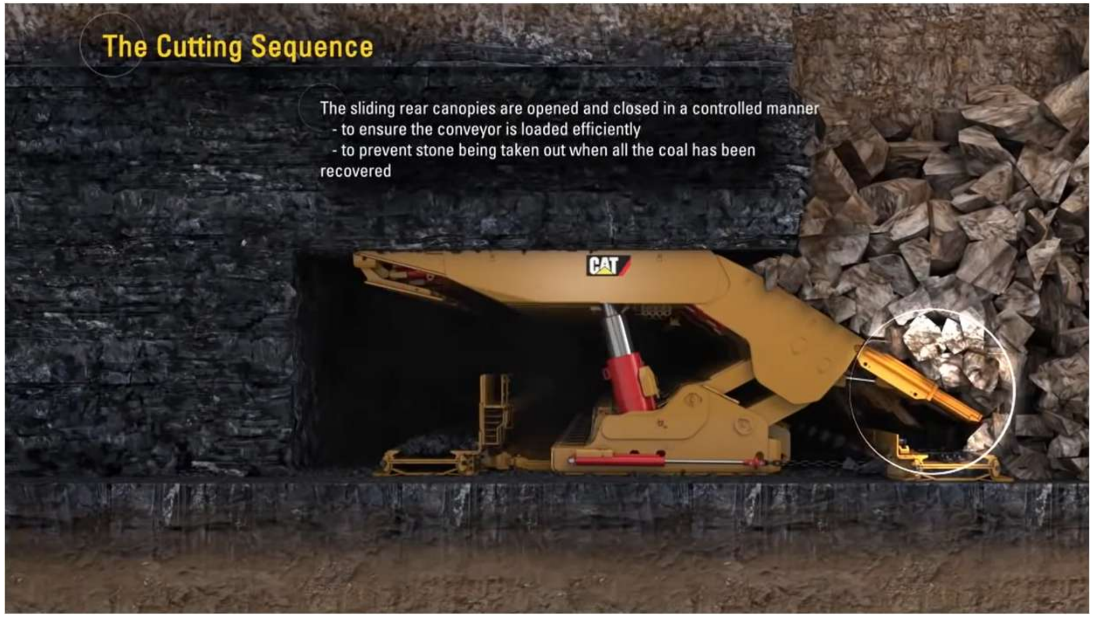


# Sub-Level Caving

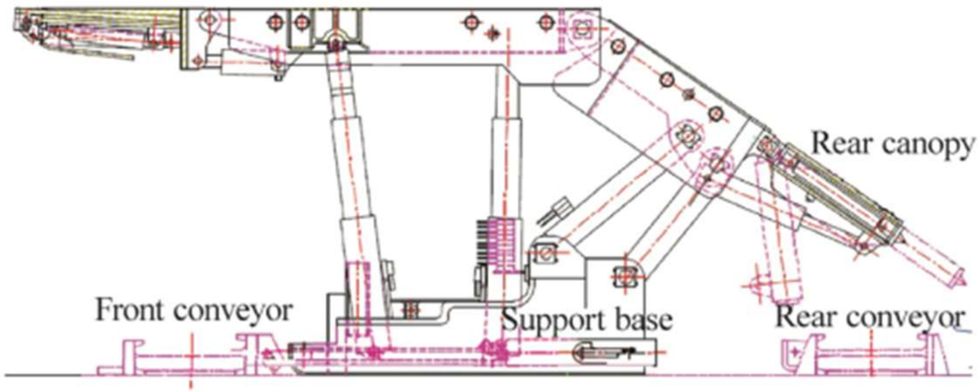



The Cutting Sequence
CAT


The Cutting Sequence
The sliding rear canopies are sequentially opened.
The coal falls through onto the rear mounted AFC.
CAT


The Cutting Sequence
The sliding rear canopies are opened and closed in a controlled manner
- to ensure the conveyor is loaded efficiently
- to prevent stone being taken out when all the coal has been recovered
CAT

# LTCC


Rear canopy
Front conveyor
Support base
Rear conveyor